Graph-Based Blind Hyperspectral Unmixing via Nonnegative Matrix Factorization

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***Abstract*— Hyperspectral unmixing (HU) is a crucial step in the hyperspectral image (HSI) analysis. It aims at decomposing the observed spectrum at each pixel into a collection of constituent endmembers, weighted by their abundances. In a close spatial neighborhood, due to spatial autocorrelation, the abundances of an endmember tend to be similar to each other. Therefore, the spectra at the neighboring pixels are also very much alike. Recently, many studies have focused on graph-regularized nonnegative matrix factorization (NMF) approaches to address the blind HU problem. This is mainly because graph structures allow spatial–spectral relations to be effectively embedded in the NMF framework. This article takes one step further by defining each abundance map as a signal on a suitable graph, which enables the HU problem to be analyzed for the first time from a graph signal processing perspective. This article introduces a novel Laplacian regularizer based on the l_1-norm, where graph spectral analysis is utilized to show that the regularizer has natural piecewise smooth (PWS) signal promotion and noise rejection capabilities. A graph-based blind HU algorithm is developed by incorporating this regularizer and an $l_{1/2}$-abundance sparsity constraint into the NMF problem. Since the featured regularizer exploits the PWS property of abundance maps, the proposed method is effective in HU. The experiments conducted on synthetic and real HSIs demonstrate that it has superior performance compared with several popular HU algorithms.**

***Index Terms*— Blind source separation, graph signal processing (GSP), hyperspectral image (HSI), HSI unmixing, nonnegative matrix factorization (NMF).**

I. Introduction

HYPERSPECTRAL images (HSIs) captured by imaging spectrometers contain hundreds of adjacent spectral bands [1]. Due to the availability of this abundant spectral information, HSIs are favorable in remote sensing applications, such as mineral exploration, precision agriculture, and environmental-disaster monitoring [2]–[7]. The spectral data provided by HSIs are often used to identify the materials that are present in a scene. However, due to the complicated ground covers and the low spatial resolution of hyperspectral imaging systems, the observed spectrum at each pixel is typically a mixture of several pure spectra (called endmembers) [8]. These mixed observations pose a serious impediment to HSI analysis for anomaly detection [9], clustering [10], and classification [11]. Therefore, hyperspectral unmixing (HU), which is decomposing an HSI into a collection of constituent endmember signatures and their relative proportions (called abundances), is an essential step in HSI analysis [12].

An HSI can either be a linear or nonlinear mixture of endmember signatures. However, due to the algorithmic simplicity and efficiency induced by the linear mixing mode (LMM) [13], it is still the most widely used HSI model for HU [14]. Furthermore, two abundance constraints—the abundance nonnegative constraint (ANC) and the abundance sum-to-one constraint (ASC)—are often adopted in linear unmixing in order to improve the unmixing performance [14].

An HSI is a collection of monochrome images belonging to a contiguous range of wavelengths. Hyperspectral sensors, commonly known as a spectrometer, can capture the same scene with hundreds of narrow spectral bands [2]. As a result, HSIs possess huge potentials to distinguish between different classes of earth resources. With the development of HSI analysis techniques introduced in the past few decades and, thereby, the major advancement in classification and unmixing performances using HSIs, many fields have effectively pursued utilizing HSIs for a variety of applications. Biotechnology, environmental monitoring, food quality testing, medical imaging/diagnosis, surveillance, and forensic science are some of the fields that have incorporated HSI analysis to attain improved performance in various tasks [4], [5]. Remote Sensing is another key field in which HSI analysis had dominated over the past decade since it has the ability to discover richer information due to the presence of multiple spectral bands representing data, as opposed to conventional tricolor images.

Over the past few decades, various linear unmixing algorithms have been proposed, including N-Finder [15], vertex component analysis (VCA) [16], minimum volume simplex analysis [17], independent component analysis (ICA) [18], [19], and nonnegative matrix factorization (NMF) [20]–[24]. Among these, both ICA and NMF are statistical approaches and consider HU as a blind source separation problem [25]. Recently, NMF has been gaining popularity in HU because

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it aims at discovering part-based representations (i.e., strictly additive components) from multivariate data, which greatly coincides with the objective of HU [22].

Utilizing an iterative algorithm, NMF can simultaneously decompose an HSI into two nonnegative matrices, called the "source matrix" and the "mixing matrix." In the optimum decomposition, the source matrix is the endmember matrix whose columns are the endmember signatures, and the mixing matrix is the abundance matrix whose rows yield the endmember abundance maps. However, NMF has multiple candidate solutions and is, therefore, inherently ill-posed [26]. To improve the uniqueness of the NMF solution, numerous constrained NMF methods have been proposed by adding appropriate constraints to the original NMF problem (see [3], [12], [14], [20]–[23], [27]–[30]).

Motivated by the sparse distribution of ground materials, sparsity constraints have often been imposed on abundances to reduce the number of different materials contained in each pixel [25]. Mathematically, regularizing with the l_0-norm should yield the sparsest solution; however, this leads to an NP-hard problem. Realizing this issue, Hoyer [27] considered a sparsity constraint based on the l_1-norm. The l_1-norm regularization problem (often called the Lasso problem [31]) had been frequently utilized in many fields, including signal processing, machine learning, computer vision, and statistics literature, where sparse representations are required. Sparse representations have proven to be effective in many applications, such as image restoration [32], face recognition [33], object recognition [34], speech classification [35], music genre classification [36], and also HU [27].

With regard to HU, then, many works such as [37], [38] focused on regularizing with l_p-norms ($0 < p < 1$) because the minimization of the l_1-sparsity constraint disagrees to satisfy the ASC. As far as only the sparsity constraints are considered, the $l_{1/2}$-sparsity constrained NMF can yield better unmixing results than those based on other l_p-norms ($0 < p < 1$) [22].

Due to spatial autocorrelation, abundances of an endmember tend to be similar to each other on neighboring pixels. Therefore, the corresponding abundance map is piecewise smooth (PWS), and the spectra observed at the neighboring pixels are very much alike. Fig. 1 graphically illustrates the PWS nature of the abundance maps with reference to a few standard real HSI data sets mentioned in Section V-D. The unmixing performance can be improved by incorporating constraints that account for these spatial–spectral characteristics into the NMF problem [3], [39].

The field of graph signal processing (GSP) can provide tools to analyze the spatial–spectral attributes contained in hyperspectral data. Deviating from classical digital signal processing, which requires a regular data domain, GSP describes the underlying structure of data via graphs, allowing the data domain to be irregular [40]. The data originating on these graphs are perceived as signal samples on graph nodes. Collectively, these samples are referred to as a graph signal. The weights associated with edges can contain correlations between the nodes connected by these edges. So far, data analysis and processing using GSP tools have been performed

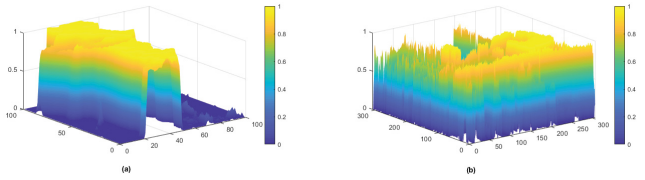


Fig. 1. 3-D visualization of the abundance values of abundance maps belonging to (a) second endmember (water) of the Jasper data set. (b) First endmember (Asphalt) of the Urban HYDICE data set, where the vertical axis represents the abundance values at each pixel location.

mostly on signal sampling and reconstruction [41], modeling and inference of diffusion processes [42], and topology identification [43]. There are also a considerable amount of work [3], [23], [30], [44]–[48], which had dealt with source separation/unmixing utilizing a graph-based unmixing perspective.

Inspired by the flexibility of graph regularization methods, this article proposes a new blind HU algorithm named the sparsity constrained graph l_1-Laplacian regularized NMF (SC-GLR$_{l_1}$NMF). The novelty of the proposed method is the incorporation of a new smoothness constraint called the graph l_1-Laplacian regularizer (GLR$_{l_1}$) into the $l_{1/2}$-sparsity constrained NMF problem. The contributions of this article are summarized as follows.

1) The development of a new blind HU algorithm named SC-GLR$_{l_1}$NMF, which effectively incorporates a novel graph-based smoothness constraint labeled GLR$_{l_1}$. This novel constraint promotes piecewise smoothness at a higher degree compared with the previously developed PWS regularization methods, which ultimately results in the proposed method outperforming those methods.
2) A GSP-based spectral analysis of the developed GLR$_{l_1}$ constraint, which provides clear evidence for its PWS signal promotion and noise rejection capabilities.

SC-GLR$_{l_1}$NMFs performance is evaluated via extensive experiments conducted on both synthetic and real HSIs. These experiments demonstrate that the unmixing performance of the proposed algorithm is superior to that of several popular HU methods.

The rest of this article is organized as follows. In Section II, the LMM, NMF, and basic GSP concepts are briefly reviewed. Section III is devoted to the derivation and analysis of GLR$_{l_1}$. The proposed HU algorithm is presented in Section IV. The experimental results are provided in Section V, and this article is concluded in Section VI.

Notation: $\mathbb{R}_+$ is the nonnegative real line. Bold uppercase letters and bold lowercase letters denote matrices and column vectors, respectively. $\mathbf{x}(i)$ is the ith element of $\mathbf{x}$, and $\mathbf{X}_{ij}$ and $\mathbf{X}_i$, respectively, are the ijthentry and the ith row of $\mathbf{X}$. The trace of $\mathbf{X}$ is represented by $\mathrm{tr}(\mathbf{X})$. A nonnegative matrix $\mathbf{X}$ is denoted by $\mathbf{X} \succeq 0$. An $n \times n$ diagonal matrix $\boldsymbol{X}$ is written as $\boldsymbol{X} = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, where λ_is are the diagonal elements.

II. Related Works

A. Linear Mixing Model

The LMM describes the spectrum at each HSI pixel as a linear combination of the constituent endmember

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signatures, weighted by the corresponding abundances. This model relies heavily on the validity of the assumption that the ground is sufficiently flat so that incident solar beams are reflected off only once, without being scattered. Let $\mathbf{X} := [\mathbf{x}_1, \ldots, \mathbf{x}_N] \in \mathbb{R}_+^{L\times N}$ be the N-pixel HSI acquired in L spectral bands, $\mathbf{A} := [\mathbf{a}_1, \ldots, \mathbf{a}_K] \in \mathbb{R}_+^{L\times K}$ be the endmember matrix that contains K endmember signatures, and $\mathbf{S} := [\mathbf{s}_1, \ldots, \mathbf{s}_N] \in \mathbb{R}_+^{K\times N}$ be the abundance matrix whose rows are the vectorized 2-D abundance maps. The column vector $\mathbf{s}_j \in \mathbb{R}^K$ concatenates endmember abundances of the jth pixel. By adopting the LMM, the spectrum recorded at the jth pixel can then be written as

$$\mathbf{x}_j = \sum_{i=1}^{K} \mathbf{S}_{ij}\mathbf{a}_i + \mathbf{n}_j = \mathbf{A}\mathbf{s}_j + \mathbf{n}_j \tag{1}$$

where $\mathbf{n}_j \in \mathbb{R}^{L\times 1}$ is the additive noise term at the same pixel. When the ANC and the ASC are imposed, we have that $\mathbf{S}_{ij} \geq 0$ and $\sum_{i=1}^{K} \mathbf{S}_{ij} = 1$, respectively. Using the representation in (1), the fabric of the HSI data $\mathbf{X}$ can be presented in matrix form as

$$\mathbf{X} = \mathbf{AS} + \mathbf{N}$$

where $\mathbf{N} = [\mathbf{n}_1, \ldots, \mathbf{n}_N] \in \mathbb{R}^{L\times N}$ is the noise matrix. Given that $\mathbf{X}$ is available, the objective of blind HU is to estimate both $\mathbf{A}$ and $\mathbf{S}$ simultaneously.

B. NMF

NMF is a linear, low-rank matrix approximation of nonnegative matrix data [49]. Its objective is to decompose a nonnegative matrix $\mathbf{X} \in \mathbb{R}_+^{L\times N}$ into a product of another two nonnegative matrices called the source matrix $\mathbf{A} \in \mathbb{R}_+^{L\times K}$ and the mixing matrix $\mathbf{S} \in \mathbb{R}_+^{K\times N}$, as

$$\mathbf{X} \approx \mathbf{AS}. \tag{2}$$

Here, each column $\mathbf{x}_n$ of $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_N]$ is approximated by a linear sum of the columns of $\mathbf{A} = [\mathbf{a}_1, \ldots, \mathbf{a}_K]$, with the weights appearing in the corresponding column of $\mathbf{S}$, i.e., $\mathbf{x}_n = \sum_{i=1}^{K} \mathbf{S}_{in}\mathbf{a}_i$. Since these weights and elements of $\mathbf{A}$ are constrained to be strictly nonnegative, NMF results in interpretable part-based representations, aptly fitting the HU framework [22]. The factorization (2) is attained via optimizing an appropriate objective function subject to the nonnegativity constraints. Using the Euclidean distance error as the objective function, the standard NMF is defined as follows.

Problem 1: Given a nonnegative matrix $\mathbf{X}$

$$\min_{\mathbf{A},\mathbf{S}} \|\mathbf{X} - \mathbf{AS}\|_F^2$$
$$\text{s.t. } \mathbf{A}, \mathbf{S} \geq 0.$$

Lee and Seung [50] provide a solution to Problem 1 using the following elementwise multiplicative update rules iteratively:

$$\mathbf{A}_{lk} \leftarrow \mathbf{A}_{lk}\frac{(\mathbf{X}\mathbf{S}^T)_{lk}}{(\mathbf{A}\mathbf{S}\mathbf{S}^T)_{lk}} \quad \mathbf{S}_{kn} \leftarrow \mathbf{S}_{kn}\frac{(\mathbf{A}^T\mathbf{X})_{kn}}{(\mathbf{A}^T\mathbf{A}\mathbf{S})_{kn}}.$$

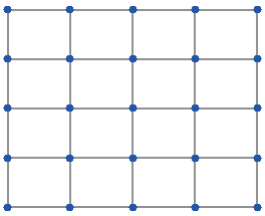


Fig. 2. Four-connectivity graph for a 5×5 image.

Here, in terms of HU, $\mathbf{A}_{lk}$ is the (lk)th element of the endmember matrix A, and $\mathbf{S}_{kn}$ is the (kn)th element of the abundance matrix S. Arriving at this solution, however, is not guaranteed due to the fact that there are infinitely many $(\mathbf{A}, \mathbf{S})$ pairs satisfying (2). For instance, we can write that $\mathbf{AS} = (\mathbf{A}\mathbf{Q}^{-1})(\mathbf{Q}\mathbf{S})$ for any invertible $\mathbf{Q} \in \mathbb{R}_+^{K\times K}$. The uniqueness of the solution can be improved by incorporating appropriate regularizers, such as sparsity and smoothness constraints into the standard NMF problem. Details of several enhanced NMF algorithms related to HU can be found in [3], [12], [14], [20]–[23], and [27]–[30].

C. GSP

By embedding the structure of a signal onto a graph, GSP extends classical signal processing to analyze and process data samples residing on irregular domains. Any vector $\mathbf{x} = [x_1, \ldots, x_n]^T$ whose elements rest on nodes of a graph $\mathcal{G}$ can be referred to as a graph signal on $\mathcal{G}$.

We can capture correlations between pixels via a graph that is connected, weighted, and undirected. To do so, we define a graph as a triplet $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{W}\}$, where $\mathcal{V} = \{1, \ldots, N\}$ is a set of N nodes representing pixels, $\mathcal{E}$ is a set of edges with (i, j) being the edge connecting the nodes $i, j \in \mathcal{V}$, and $\mathbf{W} \in \mathbb{R}_+^{N\times N}$ is the weight matrix with $\mathbf{W}_{ij} \geq 0$ being the weight of the edge (i, j). Each column of $\mathbf{S}$ is imagined to be residing on the corresponding pixel node. Then, each row of $\mathbf{S}$ can be viewed as a signal on the graph, yielding K independent graph signals. For an undirected graph $\mathcal{G}$, the weights are symmetric, i.e., $\mathbf{W}_{ij} = \mathbf{W}_{ji}$ for all (i, j). In general, the edge weight $\mathbf{W}_{ij}$ is a measure of proximity of pixels i and j, where proximity is a contextual measure. For example, in this article, proximity is quantified by spatial and spectral similarities.

Even though there are many complex graph construction methods in the literature [51], this article considers a common and straightforward construction, i.e., the four-connectivity graph, as shown in Fig. 2, for a 5×5 pixel image. The graph construction for an N pixel patch is outlined as follows. Each pixel i, $1 \leq i \leq N$ is presented as a node $i \in \mathcal{V}$. An edge $(i, j) \in \mathcal{E}$ exists between the nodes i and j if they are adjacent neighbors in the horizontal or vertical direction on the 2-D grid. Note that there are no self connections, i.e., $(i, i) \notin \mathcal{E}$. For an $\sqrt{N} \times \sqrt{N}$ node four-connectivity graph, there are $2(N - \sqrt{N})$ number of edges.

The weight $\mathbf{W}_{ij}$[1] is defined via the thresholded Gaussian kernel weighting function, using the spectra $\mathbf{x}_i$ and $\mathbf{x}_j$ at the

[1] $\mathbf{W}_{ij}$ is not to be confused with the new weight function $\mathcal{W}_{ij}$ introduced in Section III-A.

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associated pixels and a parameter $\sigma > 0$, as

$$\mathbf{W}_{ij} = \begin{cases} \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|_2^2}{\sigma^2}\right), & \text{if } (i,j) \in \mathcal{E} \\ 0. & \text{otherwise.} \end{cases} \tag{3}$$

As per (3), when the nodes i and j are neighbors in a four-connectivity graph with high spectral similarity, $\mathbf{W}_{ij}$ is large. However, when the spectra are substantially different, while the nodes are neighbors, $\mathbf{W}_{ij}$ will be comparatively smaller. Hence, in local neighborhoods, $\mathbf{W}_{ij}$ is essentially a measure of spectral similarity. Furthermore, $\mathbf{W}_{ij} = \mathbf{W}_{ji}$; hence, $\mathbf{W}$ is symmetric.

For a signal $\mathbf{x} \in \mathbb{R}^N$ on a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathbf{W}\}$, the l_2-total variation of the signal with respect to the graph is defined as [40]

$$\Phi_{l_2}(\mathbf{x}) := \frac{1}{2} \sum_{(i,j)\in\mathcal{E}} \mathbf{W}_{ij} (\mathbf{x}(i) - \mathbf{x}(j))^2.$$

Therefore, Φ_{l_2} has the following graph Laplacian quadratic (GLQ) form:

$$\Phi_{l_2}(\mathbf{x}) = \mathbf{x}^T \mathbf{L} \mathbf{x}. \tag{4}$$

Here, $\mathbf{L}$ is the Laplacian matrix associated with the graph $\mathcal{G}$. $\mathbf{L}$ is defined as the difference $\mathbf{L} := \mathbf{D} - \mathbf{W} \in \mathbb{R}^{N\times N}$,where $\mathbf{D} \in \mathbb{R}_+^{N\times N}$ is a diagonal matrix with its ith diagonal element given by the ith row sum as $\mathbf{D}_{ii} = \sum_{j=1}^{N} \mathbf{W}_{ij}$. This makes $\mathbf{L}$ real and symmetric in addition to being diagonally dominant with strictly positive diagonal elements. Therefore, $\mathbf{L}$ is positive semidefinite (PSD) [52]. With these, we can deduce an expression for $\Phi_{l_2}(\mathbf{S})$, which is the l_2-total variation of the K rows of $\mathbf{S}$ (i.e., the K abundance maps) with respect to the underlying graph. We can easily show that

$$\Phi_{l_2}(\mathbf{S}) = \frac{1}{2} \sum_{(i,j)\in\mathcal{E}} \mathbf{W}_{ij} \|\mathbf{s}_i - \mathbf{s}_j\|_2^2 \tag{5}$$

where $\mathbf{W}_{ij}$ is given by (3), and $\mathbf{s}_i$ and $\mathbf{s}_j$ are abundance vectors of the ith and jth pixels, respectively. The equivalent GLQ form is

$$\Phi_{l_2}(\mathbf{S}) = \text{tr}(\mathbf{S}\mathbf{L}\mathbf{S}^T). \tag{6}$$

This is called graph l_2-Laplacian regularizer (GLR_{l_2}) and has been used in the HU problem as a smoothness constraint, regularizing with which leads to smoothly varying abundance maps [3], [23].

In Section III, we introduce a novel smoothness constraint called the graph l_1-Laplacian regularizer (GLR_{l_1}), which has the GLQ form $\text{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T)$. The Laplacian matrix $\boldsymbol{\mathcal{L}}$ stems from a new weight matrix $\boldsymbol{\mathcal{W}}$ obtained by modifying $\mathbf{W}$. This GLR_{l_1} is an approximate measure of the l_1-total variation of $\mathbf{S}$ with respect to the graph defined as

$$\Phi_{l_1}(\mathbf{S}) := \sum_{(i,j)\in\mathcal{E}} \mathbf{W}_{ij} \|\mathbf{s}_i - \mathbf{s}_j\|_1. \tag{7}$$

The name GLR_{l_1} is due to the fact that $\text{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\boldsymbol{S}^T) \approx \Phi_{l_1}(\mathbf{S})$. We show that GLR_{l_1} has stronger PWS signal promotion capabilities compared with GLR_{l_2}, which is rather a smooth signal inducing constraint. Since abundance maps are PWS, regularizing with GLR_{l_1} is bound to yield superior unmixing performance.

Since the graph Laplacian matrix is real and symmetric, it has the following eigendecomposition:

$$\mathbf{L} = \mathbf{V}\Lambda\mathbf{V}^T \tag{8}$$

where $\Lambda = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$ is the diagonal eigenvalue matrix with $\{\lambda_n\}_{n=1,2,\ldots,N}$ being the set of eigenvalues of $\mathbf{L}$, and $\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_N]$ is the matrix of corresponding eigenvectors in the same order. Using $\mathbf{V}$, we present the graph Fourier transform (GFT) of a signal $\mathbf{x}$.

Definition 1 [40]: Let $\mathbf{x} \in \mathbb{R}^N$ be a graph signal defined on a graph characterized by a Laplacian matrix $\mathbf{L} \in \mathbb{R}^{N\times N}$ having the eigendecomposition in (8). The GFT of $\mathbf{x}$ with respect to $\mathbf{L}$ is defined as

$$\widehat{\mathbf{x}} := \mathbf{V}^T\mathbf{x}.$$

The inverse GFT (IGFT) of $\widehat{\mathbf{x}}$ with respect to $\mathbf{L}$ is defined as

$$\mathbf{x} := \mathbf{V}\widehat{\mathbf{x}}. \tag{9}$$

Then, by (9), rewriting $\widehat{\mathbf{x}}$ as $\widehat{\mathbf{x}} = [\widehat{x}_1, \ldots, \widehat{x}_N]^T$, the graph signal $\mathbf{x}$ can be represented as

$$\mathbf{x} = \sum_{n=1}^{N} \widehat{x}_n \mathbf{v}_n. \tag{10}$$

Here, the signal $\mathbf{x}$ is expressed as a linear sum of the mutually orthogonal vectors $\mathbf{v}_n$ whose contributions to the signal are given by the corresponding GFT components $\widehat{x}_n$.

III. Graph l_1-Laplacian Regularizer

Often, l_p-norm-based smoothness constraints are used for various inverse problems, such as image denoising, deblurring, and super-resolution [45]. In these problems, l_p-norm-based methods with $p \neq 2$ result in more favorable results, and different p values are desirable for different problems. In [53], the empirical success of l_p-norm-based methods are theoretically explained. We have already established the fact that due to spatial autocorrelation between the pixels of an HSI, abundances of an endmember tend to be similar to each other on neighboring pixels, and therefore, the corresponding abundance map is PWS. It is known that in situations when PWS signals are needed to be reconstructed, the empirical optimal value of p is close to 1 [54]. This can be explained by analyzing the continuous approximation of the l_p-criterion [55]. Therefore, we introduce a novel l_1-norm-based smoothness constraint, called GLR_{l_1}, which could be embedded in the NMF algorithm in order to address the HU problem.

A. Derivation of GLR_{l_1}

As given in (6), Φ_{l_2} has a GLQ form, which we call GLR_{l_2}. However, Φ_{l_1} defined in (7) does not have such an interpretation. Here, we obtain a new weight matrix $\boldsymbol{\mathcal{W}}$ such that it results in a GLQ form which is an approximation of Φ_{l_1}. The steps of the derivation are as follows.

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We begin by close inspection of the structure of Φ_{l_2}. From (5) and (6), we can write

$$\Phi_{l_2}(\mathbf{S}) = \mathrm{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T) = \frac{1}{2}\sum_{(i,j)\in\mathcal{E}} \mathbf{W}_{ij}(\mathbf{s}_i - \mathbf{s}_j)^T(\mathbf{s}_i - \mathbf{s}_j)$$

which we differentiate with respect to $\mathbf{s}_i - \mathbf{s}_j$ to obtain

$$\frac{\partial \Phi_{l_2}(\mathbf{S})}{\partial(\mathbf{s}_i - \mathbf{s}_j)} = \mathbf{W}_{ij}(\mathbf{s}_i - \mathbf{s}_j). \tag{11}$$

Analogously, if we can find $\boldsymbol{\mathcal{W}}_{ij} \geq 0$ such that

$$\frac{\partial \Phi_{l_1}(\mathbf{S})}{\partial(\mathbf{s}_i - \mathbf{s}_j)} = \boldsymbol{\mathcal{W}}_{ij}(\mathbf{s}_i - \mathbf{s}_j) \tag{12}$$

then it is possible to present Φ_{l_1} in the GLQ form $\mathrm{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T)$ as Φ_{l_2} in (6). The matrix $\boldsymbol{\mathcal{L}}$ is a new Laplacian matrix associated with $\boldsymbol{\mathcal{W}}$. However, later in this section, we show that the set of equalities in (12) holds only approximately in an optimal sense for $\boldsymbol{\mathcal{W}}_{ij}$.

Unlike Φ_{l_2}, the function Φ_{l_1} is a nondifferentiable convex function of $\mathbf{s}_i - \mathbf{s}_j$. Therefore, we will set off to find an appropriate subgradient as below. We can rewrite $\Phi_{l_1}(\mathbf{S})$ as

$$\Phi_{l_1}(\mathbf{S}) = \sum_{(i,j)\in\mathcal{E}} \mathbf{W}_{ij}\mathbf{z}_{ij}^T(\mathbf{s}_i - \mathbf{s}_j) \tag{13}$$

for an $\mathbf{z}_{ij} \in \mathbb{R}^K$, where each element of $\mathbf{z}_{ij}$ is given by

$$\mathbf{z}_{ij}(k) = \begin{cases} 1, & \text{if } \mathbf{s}_i(k) - \mathbf{s}_j(k) > 0 \\ -1, & \text{if } \mathbf{s}_i(k) - \mathbf{s}_j(k) < 0 \\ 1 \text{ or } -1. & \text{if } \mathbf{s}_i(k) - \mathbf{s}_j(k) = 0. \end{cases} \tag{14}$$

The form in (13) is differentiable in $\mathbf{s}_i - \mathbf{s}_j$ and has a unique subgradient $\mathbf{W}_{ij}\mathbf{z}_{ij}$. Thus, for $\|\mathbf{s}_i - \mathbf{s}_j\|_1 \neq 0$, we can write

$$\frac{\partial \Phi_{l_1}(\mathbf{S})}{\partial(\mathbf{s}_i - \mathbf{s}_j)} = \mathbf{W}_{ij}\mathbf{z}_{ij}.$$

Recalling that $\mathbf{z}_{ij}^T(\mathbf{s}_i - \mathbf{s}_j) = \|\mathbf{s}_i - \mathbf{s}_j\|_1$, we have

$$\frac{\partial \Phi_{l_1}(\mathbf{S})}{\partial(\mathbf{s}_i - \mathbf{s}_j)} = \frac{\mathbf{W}_{ij}\mathbf{z}_{ij}\mathbf{z}_{ij}^T(\mathbf{s}_i - \mathbf{s}_j)}{\|\mathbf{s}_i - \mathbf{s}_j\|_1}. \tag{15}$$

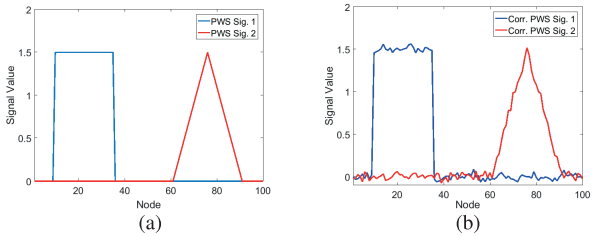
If we can find a $\beta_{ij} \in \mathbb{R}_+$, at least, such that $\mathbf{z}_{ij}\mathbf{z}_{ij}^T(\mathbf{s}_i - \mathbf{s}_j) \approx \beta_{ij}(\mathbf{s}_i - \mathbf{s}_j)$, then (15) approximately reduces to the desired form (12). We can find such a β_{ij} by solving the following optimization problem, where $\boldsymbol{\mathcal{Z}}_{(ij)} = \mathbf{z}_{ij}\mathbf{z}_{ij}^T$ and $\boldsymbol{\delta}_{ij} = \mathbf{s}_i - \mathbf{s}_j$.

Problem 2 $\min_{\beta_{ij}} \left(\|\boldsymbol{\mathcal{Z}}_{(ij)}\boldsymbol{\delta}_{ij} - \beta_{ij}\boldsymbol{\delta}_{ij}\|_2^2\right)$*:* Let the associated objective function be denoted by $\mathbb{I}$. Then, we can obtain that

$$\begin{aligned}\mathbb{I} &= \|\boldsymbol{\mathcal{Z}}_{(ij)}\boldsymbol{\delta}_{ij} - \beta_{ij}\boldsymbol{\delta}_{ij}\|_2^2 \\ &= \left(\boldsymbol{\delta}_{ij}^T\boldsymbol{\mathcal{Z}}_{(ij)} - \beta_{ij}\boldsymbol{\delta}_{ij}^T\right)\left(\boldsymbol{\mathcal{Z}}_{(ij)}\boldsymbol{\delta}_{ij} - \beta_{ij}\boldsymbol{\delta}_{ij}\right) \\ &= \boldsymbol{\delta}_{ij}^T\boldsymbol{\mathcal{Z}}_{(ij)}^2\boldsymbol{\delta}_{ij} - 2\beta_{ij}\boldsymbol{\delta}_{ij}^T\boldsymbol{\mathcal{Z}}_{(ij)}\boldsymbol{\delta}_{ij} + \beta_{ij}^2\boldsymbol{\delta}_{ij}^T\boldsymbol{\delta}_{ij}.\end{aligned}$$

It is easy to observe that $\mathbb{I}$ is convex in β_{ij}. Hence, the function $\mathbb{I}$ has a global minimum at

$$\beta_{ij} = \frac{\boldsymbol{\delta}_{ij}^T\boldsymbol{\mathcal{Z}}_{(ij)}\boldsymbol{\delta}_{ij}}{\boldsymbol{\delta}_{ij}^T\boldsymbol{\delta}_{ij}} = \frac{\boldsymbol{\delta}_{ij}^T\mathbf{z}_{ij}\mathbf{z}_{ij}^T\boldsymbol{\delta}_{ij}}{\boldsymbol{\delta}_{ij}^T\boldsymbol{\delta}_{ij}} = \left(\frac{\|\mathbf{s}_i - \mathbf{s}_j\|_1}{\|\mathbf{s}_i - \mathbf{s}_j\|_2}\right)^2. \tag{16}$$



(a) (b)

Fig. 3. Two PWS signals on a 100-node line graph. (a) Pure signals. (b) Signals corrupted by 30-dB zero-mean white Gaussian noise.

When $\|\mathbf{s}_i - \mathbf{s}_j\|_1 \neq 0$ or equivalently $\|\mathbf{s}_i - \mathbf{s}_j\|_2 \neq 0$, it can be written from (15) that

$$\frac{\partial \Phi_{l_1}(\mathbf{S}, \mathbf{W})}{\partial(\mathbf{s}_i - \mathbf{s}_j)} \approx \frac{\mathbf{W}_{ij}\|\mathbf{s}_i - \mathbf{s}_j\|_1}{\|\mathbf{s}_i - \mathbf{s}_j\|_2^2}(\mathbf{s}_i - \mathbf{s}_j). \tag{17}$$

Note that β_{ij} given by (16) is positive.

Remark 1: More precisely, (17) can be written as

$$\frac{\partial \Phi_{l_1}(\mathbf{S}, \mathbf{W})}{\partial(\mathbf{s}_i - \mathbf{s}_j)} = \frac{\mathbf{W}_{ij}\|\mathbf{s}_i - \mathbf{s}_j\|_1}{\|\mathbf{s}_i - \mathbf{s}_j\|_2^2}(\mathbf{s}_i - \mathbf{s}_j) + \mathbf{r}$$

with $\mathbf{r} \in \mathbb{R}^K$ being the residual term. The vector $\mathbf{r}$ is independent of $\mathbf{s}_i - \mathbf{s}_j$ because solving Probelm 2, in other words, means finding the projection of the vector $\mathbf{z}_{ij}\mathbf{z}_{ij}^T(\mathbf{s}_i - \mathbf{s}_j)$ along $\mathbf{s}_i - \mathbf{s}_j$. When β_{ij} is given by (16), then $\mathbf{r}$ should be orthogonal to $\mathbf{s}_i - \mathbf{s}_j$.

At this point, we regard $\mathbf{W}_{ij}\|\mathbf{s}_i - \mathbf{s}_j\|_1/\|\mathbf{s}_i - \mathbf{s}_j\|_2^2$ as the ijth element of the new weight matrix $\boldsymbol{\mathcal{W}}$, where $(i, j) \in \mathcal{E}$. We set $\boldsymbol{\mathcal{W}}_{ij} = 0$ if $(i, j) \notin \mathcal{E}$. Along with the modifications done to ensure the numerical stability around $\|\mathbf{s}_i - \mathbf{s}_j\|_1 = \|\mathbf{s}_i - \mathbf{s}_j\|_2 = 0$, in the following, we present $\boldsymbol{\mathcal{W}}_{ij}$ for all $i, j \in \mathcal{V}$

$$\boldsymbol{\mathcal{W}}_{ij} = \begin{cases} \mathbf{W}_{ij}\dfrac{\|\mathbf{s}_i - \mathbf{s}_j\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_i - \mathbf{s}_j\|_2^2 + \epsilon^2}, & \text{if } (i, j) \in \mathcal{E} \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

Here, ϵ is a small positive constant, and in the experiments, we fix it at 0.001. Note that the associated weight matrix $\boldsymbol{\mathcal{W}}$ is real and symmetric.

From the Cauchy–Schwarz inequality, it can easily be shown that $\|\mathbf{s}_i - \mathbf{s}_j\|_1 \leq \sqrt{K}\|\mathbf{s}_i - \mathbf{s}_j\|_2$. Therefore, when $\|\mathbf{s}_i - \mathbf{s}_j\|_1 \gg \sqrt{K}\epsilon$

$$\boldsymbol{\mathcal{W}}_{ij} \approx \mathbf{W}_{ij}\frac{\|\mathbf{s}_i - \mathbf{s}_j\|_1}{\|\mathbf{s}_i - \mathbf{s}_j\|_2^2}$$

whereas when $\|\mathbf{s}_i - \mathbf{s}_j\|_2 \ll \epsilon$

$$\boldsymbol{\mathcal{W}}_{ij} \approx \frac{\sqrt{K}}{\epsilon}\mathbf{W}_{ij}.$$

When $\|\mathbf{s}_i - \mathbf{s}_j\|_2 = \|\mathbf{s}_i - \mathbf{s}_j\|_1 = 0$, the weight $\boldsymbol{\mathcal{W}}_{ij}$ is given by $\boldsymbol{\mathcal{W}}_{ij} = \sqrt{K}\mathbf{W}_{ij}/\epsilon$ and is, therefore, large. At this instance, this is highly desirable as the weight $\boldsymbol{\mathcal{W}}_{ij}$ is required to reflect the high correlation between the nodes i and j.

A Laplacian matrix $\boldsymbol{\mathcal{L}}$ is calculated as $\boldsymbol{\mathcal{L}} = \boldsymbol{\mathcal{D}} - \boldsymbol{\mathcal{W}}$, where $\boldsymbol{\mathcal{D}}$ is the diagonal degree matrix associated with $\boldsymbol{\mathcal{W}}$. Since $\boldsymbol{\mathcal{L}}$ is real and symmetric, it has a similar eigendecomposition as the conventional Laplacian matrix $\mathbf{L}$, as described in Section II-C.

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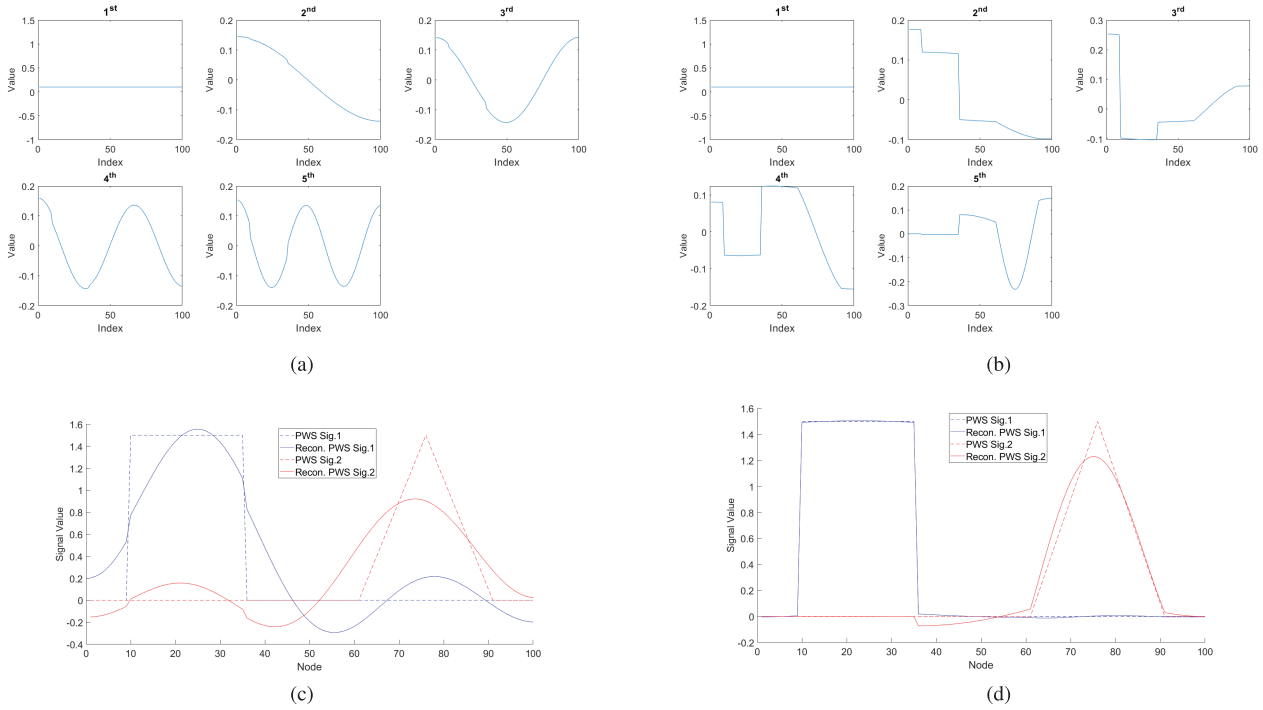


Fig. 4. Eigenvectors of $\mathbf{L}$ and $\boldsymbol{\mathcal{L}}$ associated with the pure graph signals in Fig. 3(a). (a) First five eigenvectors of $\mathbf{L}$. (b) First five eigenvectors of $\boldsymbol{\mathcal{L}}$. (c) Reconstruction of the pure signals with the first five eigenvectors of $\mathbf{L}$. (d) Reconstruction of the pure signals with the first five eigenvectors of $\boldsymbol{\mathcal{L}}$.

The matrix $\boldsymbol{\mathcal{L}}$ is used to obtain the GLQ form $\text{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T)$, which is theoretically the best GLQ approximation of Φ_{l_1} and, hence, the name GLR_{l_1}.

Next, we will explore the spectral properties of the Laplacian matrix $\boldsymbol{\mathcal{L}}$ and the effect of GLR_{l_1} on HU.

B. Spectral Analysis of GLR_{l_1}

For the convenience of demonstrating the spectral properties of L, we consider two PWS signals residing on a line graph (see Fig. 3). An edge (in the graph sense) between two nodes exists if and only if these nodes are immediate neighbors in the line graph. In the HSI setting, the considered graph signals are analogous to two 1-D abundance maps. Thus, these two PWS signals were multiplied by a random mixing matrix in order to generate the mixed pixels or the HSI.

Fig. 4(a) and (b), respectively, shows the first five eigenvectors of the Laplacian matrices $\mathbf{L}$ and $\boldsymbol{\mathcal{L}}$ associated with the pure graph signals shown in Fig. 3(a). For the calculation of $\mathbf{L}$, the $\mathbf{W}$ matrix is computed using (3). For the calculation of $\boldsymbol{\mathcal{L}}$, the $\boldsymbol{\mathcal{W}}$ matrix is computed using (18), and also for the calculation of $\boldsymbol{\mathcal{W}}$, the $\mathbf{W}$ matrix is computed using (3).

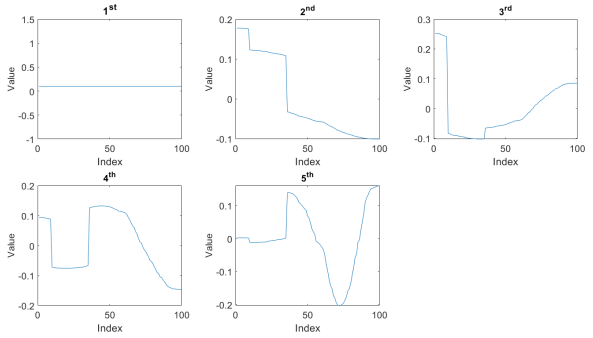
The shown eigenvectors correspond to the smallest five eigenvalues of the respective Laplacian matrices. It can be observed that the components of the eigenvectors of $\boldsymbol{\mathcal{L}}$ vary in a PWS manner. The eigenvectors of $\mathbf{L}$, on the other hand, are merely smooth signals. The reconstructions of the two PWS signals with the lowest five eigenvectors of $\mathbf{L}$ and $\boldsymbol{\mathcal{L}}$ are shown in Fig. 4(c) and (d), respectively [these follow from IGFT (9) and (10)]. The lower eigenvectors of $\boldsymbol{\mathcal{L}}$ can construct PWS signals due to their PWS properties, whereas those of $\mathbf{L}$ are not that effective to this respect.

The spectral properties of $\boldsymbol{\mathcal{L}}$ can be mathematically interpreted by further analyzing the behavior of the l_1 and l_2-norms. The l_2-norm being a differentiable norm in $\mathbb{R}^N$ (everywhere except at the origin) is smooth in $\mathbb{R}^N$ [56]. In contrast, the l_1-norm is not differentiable everywhere in $\mathbb{R}^N$. It is differentiable piecewise and, therefore, is PWS in $\mathbb{R}^N$. Piecewise smoothness can be a consequence of either discontinuities or sharp edges in functions. In the HSI domain, both these causes are possible, but the sharp edges are predominantly common. Constructing a neighborhood of a PWS function requires an infinite sum of smooth functions. Conversely, such a neighborhood may be constructed using an optimally selected finite set of PWS functions. In the example discussed earlier, the continuous extension of the eigenvectors of $\mathbf{L}$ is smooth, whereas the continuous extension of the eigenvectors of $\boldsymbol{\mathcal{L}}$ is PWS, thereby can accommodate discontinuities or sharp edges. As a result, (19) will accommodate PWS eigenvectors $\mathbf{v}_n$ resulting in PWS abundance maps $\mathbf{S}_m$.

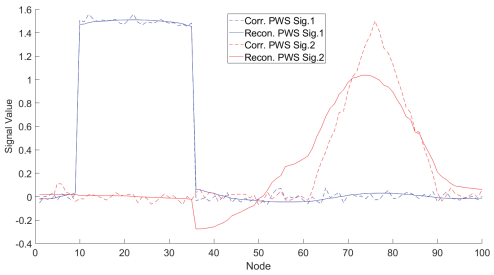
Fig. 3(b) shows the same two PWS signals when corrupted by 30-dB zero-mean white Gaussian noise. The first five eigenvectors of the corresponding $\boldsymbol{\mathcal{L}}$ are shown in Fig. 5(a). Despite the noisy signals, these eigenvectors are still capable of maintaining the piecewise smoothness. Fig. 5(b) demonstrates the reconstruction of the signals with these eigenvectors, and we observe that they are sufficiently PWS and less noisy.

Next, we will see the effect of GLR_{l_1} on the HU problem. Toward this, we consider a less restricted problem, i.e., semi-blind HU, where we seek to extract the abundance matrix $\mathbf{S}$ from the observed HSI $\mathbf{X}$, assuming that the contributing endmember signatures come from a known library, i.e., the endmember matrix $\mathbf{A}$ is known up to a library. We show

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(a)



(b)

Fig. 5. Eigenvectors of $\boldsymbol{\mathcal{L}}$ associated with the corrupted graph signals in Fig. 3(b). (a) First five eigenvectors of $\boldsymbol{\mathcal{L}}$. (b) Reconstruction of the corrupted graph signals with the first five eigenvectors of $\boldsymbol{\mathcal{L}}$.

under weak assumptions that this problem has an approximate closed-form solution to $\mathbf{S}$, which is directly related to the eigenvectors of the underlying graph (i.e., graph spectral bases). Let us consider the following optimization problem.

Problem 3: $\min_{\mathbf{S}} \left(\|\mathbf{X} - \mathbf{AS}\|_F^2 + \gamma \ \mathrm{tr}\left(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T\right)\right)$.

Here, γ is a positive parameter establishing a tradeoff between the importance of the reconstruction accuracy and the smoothness rendered by GLR_{l_1}. By simply substituting $\mathbf{L}$ for $\boldsymbol{\mathcal{L}}$ in Problem 3, one can change the smoothness prior to GLR_{l_2}. It can be shown that when $\mathbf{A}^T\mathbf{A}$ and $-\gamma\,\boldsymbol{\mathcal{L}}$ do not have common eigenvalues, the rowwise approximate solution to Problem 3 is given by (see Appendix A for a detailed derivation)

$$\mathbf{S}_m \approx \sum_{n=1}^{N} \left(\sum_{k=1}^{K} \mathbf{U}_{mk}\mathbf{C}_{kn} \frac{\mathbf{v}_n^T}{\mu_k + \gamma\,\lambda_n} \right). \tag{19}$$

Here, $\mathbf{C} = \mathbf{U}^T(\mathbf{A}^T\mathbf{X})\mathbf{V}$, where $\mathbf{U}$ and $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_N]$ are such that

$$\mathbf{U}^T(\mathbf{A}^T\mathbf{A})\mathbf{U} = \mathrm{diag}(\mu_1, \ldots, \mu_K) \tag{20}$$

and

$$\mathbf{V}^T\boldsymbol{\mathcal{L}}\mathbf{V} = \mathrm{diag}(\lambda_1, \ldots, \lambda_N). \tag{21}$$

Equations (20) and (21) are the eigendecompositions of $\mathbf{A}^T\mathbf{A}$ and $\boldsymbol{\mathcal{L}}$, respectively. This solution is exact for the case when Problem 3 involves GLR_{l_2}. Each recovered signal given by (19) is a linear combination of $\mathbf{v}_n^T, n = 1, \ldots, N$, i.e., the eigenvectors of $\boldsymbol{\mathcal{L}}$. We can observe that the contribution of the eigenvectors associated with larger eigenvalues is small because each $\mathbf{v}_n^T$ is weighted by a term $\sum_{k=1}^{K} \mathbf{U}_{mk}\mathbf{C}_{kn}/(\mu_k + \gamma\,\lambda_n)$. This is expected as the eigenvectors corresponding to larger eigenvalues vary rapidly with respect to the graph; thus, they are unlikely to produce smooth signals. The influence of the eigenvectors associated with smaller eigenvalues, on the other hand, is prominent in (19). Since the lower eigenvectors of $\boldsymbol{\mathcal{L}}$ are naturally PWS, as shown in the above-mentioned case study, GLR_{l_1} is bound to yield PWS signals. Furthermore, due to their noise suppression capability, GLR_{l_1} should be able to recover PWS signals even when the observed HSI $\mathbf{X}$ is noisy.

IV. Sparsity Constrained and Graph l_1-Laplacian Regularized NMF

In this section, we describe the blind HU algorithm that incorporates GLR_{l_1} into the NMF unmixing framework. In addition to this PWS signal inducing prior, we use the $l_{1/2}$-sparsity constraint to further improve the uniqueness of unmixing results, hence the name SC-GLR_{l_1}NMF.

A. SC-GLR_{l_1}NMF

We formally present the SC-GLR_{l_1}NMF problem as follows.

Problem 4:

$$\min_{\mathbf{A},\mathbf{S}} \left(\|\mathbf{X} - \mathbf{AS}\|_F^2 + \rho\|\mathbf{S}\|_{\frac{1}{2}} + \gamma \ \mathrm{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T) \right)$$
$$\text{s.t } \mathbf{A}, \mathbf{S} \succeq 0.$$

Here, $\|\mathbf{S}\|_{1/2} = \sum_{\forall i,j} \mathbf{S}_{ij}^{\frac{1}{2}}$ is the $l_{1/2}$-sparsity measure, and ρ and γ are two positive parameters that control the tradeoff between the reconstruction accuracy, sparseness, and the piecewise smoothness. If the objective function associated with Problem 4 is denoted by $\mathbb{J}$, then we have that

$$\mathbb{J} = \|\mathbf{X} - \mathbf{AS}\|_F^2 + \rho\|\mathbf{S}\|_{\frac{1}{2}} + \gamma \ \mathrm{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T).$$

It is easy to realize that $\mathbb{J}$ is not convex in $\mathbf{A}$ and $\mathbf{S}$ simultaneously. Therefore, any algorithmic solution to $\mathbf{A}$ and $\mathbf{S}$ cannot guarantee that $\mathbb{J}$ would converge to the global minimum. Using the method of the Lagrange multipliers, we obtain multiplicative update rules for $\mathbf{A}$ and $\mathbf{S}$ for each iteration, which can, instead, steer $\mathbb{J}$ to a local minimum.

As the minimization of $\mathbb{J}$ is subject to the constraints $\mathbf{A}_{lk} \geq 0$ and $\mathbf{S}_{kn} \geq 0$, the corresponding Lagrange multipliers are defined as $\Psi_{lk} \geq 0$ and $\Phi_{kn} \geq 0$. If the Lagrange function is denoted by $\mathbb{L}$, then

$$\begin{aligned}\mathbb{L} &= \|\mathbf{X} - \mathbf{AS}\|_F^2 + \rho\|\mathbf{S}\|_{\frac{1}{2}} + \gamma \ \mathrm{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T) \\ &\quad + \mathrm{tr}(\Psi\mathbf{A}^T) + \mathrm{tr}(\Phi\mathbf{S}^T) \\ &= \mathrm{tr}(\mathbf{X}^T\mathbf{X}) + \mathrm{tr}(\mathbf{S}^T\mathbf{A}^T\mathbf{AS}) - 2\,\mathrm{tr}(\mathbf{X}^T\mathbf{AS}) \\ &\quad + \rho\|\mathbf{S}\|_{\frac{1}{2}} + \gamma \ \mathrm{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T) + \mathrm{tr}(\Psi\mathbf{A}^T) + \mathrm{tr}(\Phi\mathbf{S}^T).\end{aligned}$$

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Taking partial derivative of both sides with respect to $\mathbf{A}_{lk}$ and $\mathbf{S}_{kn}$, respectively, yields

$$\frac{\partial \mathbb{L}}{\partial \mathbf{A}_{lk}} = 2(\mathbf{A}\mathbf{S}\mathbf{S}^T)_{lk} - 2(\mathbf{X}\mathbf{S}^T)_{lk} + \Psi_{lk}$$
$$\frac{\partial \mathcal{L}}{\partial \mathbf{S}_{kn}} = 2(\mathbf{A}^T\mathbf{A}\mathbf{S})_{kn} - 2(\mathbf{A}^T\mathbf{X})_{kn} + \frac{1}{2}\rho(\mathbf{S}_{kn})^{-\frac{1}{2}} + \gamma\frac{\partial \operatorname{tr}(\mathbf{S}\mathcal{L}\mathbf{S}^T)}{\partial \mathbf{S}_{kn}} + \Phi_{kn}.$$

If $\operatorname{tr}(\mathbf{S}\mathcal{W}\mathbf{S}^T) = F$ and $\operatorname{tr}(\mathbf{S}\mathcal{D}\mathbf{S}^T) = G$, then, as $\mathcal{L} = \mathcal{D} - \mathcal{W}$, we get

$$\frac{\partial \operatorname{tr}(\mathbf{S}\mathcal{L}\mathbf{S}^T)}{\partial \mathbf{S}_{kn}} = \frac{\partial G}{\partial \mathbf{S}_{kn}} - \frac{\partial F}{\partial \mathbf{S}_{kn}}.$$

Assuming that abundances on neighboring pixels are similar, we can show that

$$\frac{\partial F}{\partial \mathbf{S}_{kn}} = 2(\mathbf{S}\mathcal{W})_{kn} + \sum_{\substack{m \text{ s.t.} \\ (m,n)\in\mathcal{E}}} \frac{\big(\mathbf{z}_{nm}(k) - \mathbf{z}_{mn}(k)\big)\mathbf{W}_{nm}}{\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2}(\mathbf{S}^T\mathbf{S})_{nm} \tag{22}$$

$$\frac{\partial G}{\partial \mathbf{S}_{kn}} = 2(\mathbf{S}\mathcal{D})_{kn} + \sum_{\substack{i,p \text{ s.t.} \\ (n,p)\in\mathcal{E}}} \big(\mathbf{z}_{np}(k)\mathbf{S}_{in}^2 - \mathbf{z}_{pn}(k)\mathbf{S}_{ip}^2\big)\frac{\mathbf{W}_{np}}{\|\mathbf{s}_n - \mathbf{s}_p\|_2^2 + \epsilon^2}. \tag{23}$$

The term $\mathbf{z}_{ij}(k)$ is defined in (14). For details of the derivation, see Appendix B.

With the Karun Kush Tucker (KKT) conditions $\Psi_{lk}\mathbf{A}_{lk} = 0$ and $\Phi_{kn}\mathbf{S}_{kn} = 0$, we can obtain the following formulas for $\mathbf{A}_{lk}$ and $\mathbf{S}_{kn}$:

$$(\mathbf{A}\mathbf{S}\mathbf{S}^T)_{lk}\mathbf{A}_{lk} - (\mathbf{X}\mathbf{S}^T)_{lk}\mathbf{A}_{lk} = 0$$
$$(\mathbf{A}^T\mathbf{A}\mathbf{S})_{kn}\mathbf{S}_{kn} - (\mathbf{A}^T\mathbf{X})_{kn}\mathbf{S}_{kn} + \frac{\rho}{4}(\mathbf{S}_{kn})^{-\frac{1}{2}}\mathbf{S}_{kn} + \frac{\gamma}{2}\frac{\partial G}{\partial \mathbf{S}_{kn}}\mathbf{S}_{kn} - \frac{\gamma}{2}\frac{\partial F}{\partial \mathbf{S}_{kn}}\mathbf{S}_{kn} = 0.$$

These equations lead to the following multiplicative update rules:

$$\mathbf{A}_{lk} \leftarrow \mathbf{A}_{lk}\frac{(\mathbf{X}\mathbf{S}^T)_{lk}}{(\mathbf{A}\mathbf{S}\mathbf{S}^T)_{lk}} \tag{24}$$

$$\mathbf{S}_{kn} \leftarrow \mathbf{S}_{kn}\frac{(\mathbf{A}^T\mathbf{X})_{kn} + \frac{\gamma}{2}\frac{\partial F}{\partial \mathbf{S}_{kn}}}{(\mathbf{A}^T\mathbf{A}\mathbf{S})_{kn} + \frac{\rho}{4}(\mathbf{S}_{kn})^{-\frac{1}{2}} + \frac{\gamma}{2}\frac{\partial G}{\partial \mathbf{S}_{kn}}}. \tag{25}$$

Next, we will discuss several issues associated with the implementation of the SC-GLR$_{l_1}$NMF algorithm.

B. Algorithm Implementation Issues

The first issue is how to determine the number of endmembers in an HSI scene, for which there are several effective methods, such as virtual dimensionality [57] and HySime [58]. However, it is another independent topic and is outside the scope of this article. Therefore, we assume that the number of endmembers in an HSI is already known.

The second issue is associated with the algorithm initialization and termination. In our implementation, the matrix $\mathbf{A}$ is initialized using the VCA algorithm [16], and $\mathbf{S}$ is initialized using the fully constrained least squares (FCLS) algorithm [59]. The algorithm is terminated either if the number of iterations t reaches a preset maximum value ($T_{\max}$) or if the condition $|\mathcal{J}^{(t)} - \mathcal{J}^{(t-1)}|/\mathcal{J}^{(t-1)} \le 1 \times 10^{-3}$ is met. Here, $\mathcal{J}^{(t)}$ is the reconstruction error $\|\mathbf{X} - \mathbf{A}^{(t)}\mathbf{S}^{(t)}\|_2$ at the tth iteration.

The third issue is imposing the ASC. Adopting the method proposed in [59], both $\mathbf{X}$ and $\mathbf{A}$ are augmented as

$$\mathbf{X}_a \leftarrow \begin{bmatrix}\mathbf{X} \\ \delta\mathbf{1}_N^T\end{bmatrix}, \quad \mathbf{A}_a \leftarrow \begin{bmatrix}\mathbf{A} \\ \delta\mathbf{1}_K^T\end{bmatrix}. \tag{26}$$

Then, $\mathbf{X}$ and $\mathbf{A}$ in (25) are, respectively, replaced by $\mathbf{X}_a$ and $\mathbf{A}_a$ in each iteration. $\mathbf{1}_N$ and $\mathbf{1}_K$ are all-one column vectors with dimensions N and K, respectively. The term δ in (26) is a positive value used to tradeoff the estimate accuracy and the ASC. The columnwise sum of $\mathbf{S}$ is driven toward unity as δ increases [14]. In the experiments, we set δ to 15.

The fourth issue is associated with the $l_{1/2}$-sparsity inducing term in the update rule (25). The operation $\mathbf{S}_{ij}^{-1/2}$ is invalid when $\mathbf{S}_{ij} = 0$. Thus, following the same procedure as described in [1], we modify the matrix $\mathbf{S}$ obtained in each update step (25) such that every entry of $\mathbf{S}$ is given by

$$\mathbf{S}_{ij} = \begin{cases}\mathbf{S}_{ij}, & \text{if } \mathbf{S}_{ij} \ge 1 \times 10^{-4} \\ 10^{-4}, & \text{if } \mathbf{S}_{ij} < 1 \times 10^{-4}.\end{cases}$$

Finally, we are faced with the issue of selecting the parameters of the proposed SC-GLR$_{l_1}$NMF. Four parameters are required to be set in advance as inputs to the proposed algorithm: σ in the Gaussian kernel weighting function (3), the regularization parameters ρ and γ in Problem 4, and $T_{\max}$ in the stopping criterion. We simply set σ to 1 because $\|\mathbf{x}_i - \mathbf{x}_j\|_2$ is itself sufficient to impart the desired properties to $\mathbf{W}_{ij}$ in (3). A proper value for the regularization parameter ρ that controls the sparseness of the solution to $\mathbf{S}$ can be determined as follows [27]:

$$\rho = \frac{1}{\sqrt{L}}\sum_{l=1}^{L}\frac{\sqrt{N} - \|\mathbf{X}_l\|_1/\|\mathbf{X}_l\|_2}{\sqrt{N} - 1}. \tag{27}$$

As for how to select appropriate values for γ and $T_{\max}$, it will be discussed in Section V-A. The proposed SC-GLR$_{l_1}$NMF approach is listed in Algorithm 1.

V. Experiments and Discussion

A. Experimental Setting

In this section, several experiments are conducted on synthetic and real-world HSI data in order to evaluate the performance of the proposed HU algorithm, i.e., SC-GLR$_{l_1}$NMF. We compare the performance of our algorithm with five popular blind HU algorithms: $l_{1/2}$-NMF [22], total variation reweighted sparse NMF (TV-RSNMF) [12], graph regularized $L_{1/2}$-NMF (GLNMF) [23] (based on GLR$_{l_2}$), HG$L_{1/2}$-NMF [3], and Spatial Group Sparsity-regularized NMF (SGSNMF) [14].

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Algorithm 1 SC-GLR$_{l_1}$NMF Algorithm for HU

Input: The HSI $\mathbf{X}$, the number of endmembers K, σ in (3), the regularization parameter γ in *Problem 4*, and the maximum number of iterations $T_{\max}$.

Output: The estimated $\mathbf{A}$ and $\mathbf{S}$.

Compute ρ from (27);
Initialize $\mathbf{A}$ using the VCA algorithm [16] and initialize $\mathbf{S}$ using the FCLS algorithm [59];
Compute $\mathbf{W}$ from (3);
Determine $\mathcal{W}$ from (18) and the corresponding $\mathcal{D}$;
Augment $\mathbf{X}$ to generate $\mathbf{X}_a$ as in (26);
repeat
  Update $\mathbf{A}$ according to (24);
  Augment $\mathbf{A}$ to generate $\mathbf{A}_a$ as in (26);
  Update $\mathbf{S}$ according to (25) with $\mathbf{X}$ and $\mathbf{A}$ replaced by $\mathbf{X}_a$ and $\mathbf{A}_a$, respectively;
  Determine $\mathcal{W}$ from (18) and the corresponding $\mathcal{D}$;
until the stopping criterion of the algorithm is satisfied.

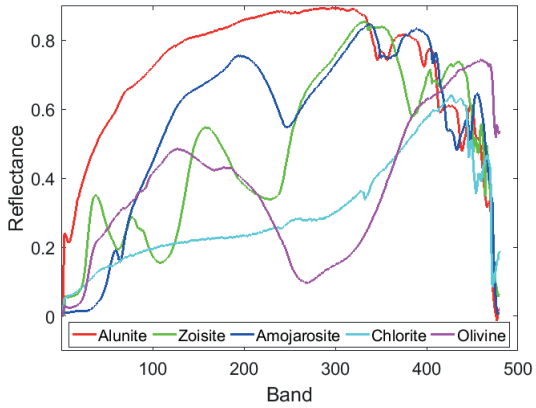


Fig. 6. Endmember signatures from the USGS library, which are used to create the simulated data set.

A simulated data set, with a size of 64 × 64 pixels and 480 spectral channels, was generated using the Hyperspectral Imagery Synthesis toolbox (HSIST)[2] for MATLAB. This toolbox includes the full U.S. Geological Survey (USGS) Spectral library and allows the user to choose appropriate simulation parameters, such as the spatial dimensions, the distribution of the abundances, and the number of endmembers. In the generation of the data set, five endmembers were randomly selected, and their corresponding abundance maps were created according to the spherical Gaussian field. Fig. 6 shows the signatures of the selected endmembers: "1# Alunite," "2# Zoisite," "3# Amojarosite," "4# Chlorite," and "5# Olivine." The corresponding abundance maps are shown in the top row of Fig. 9(a).

Regarding the selection of values for σ and ρ, we set σ to 1 and calculate ρ using (27). Here, we conduct two experiments to obtain appropriate values for γ and $T_{\max}$. Fig. 7 shows the variation of the unmixing performance of SC-GLR$_{l_1}$NMF with γ. Note that spectral angle distance

[2]http://www.ehu.es/ccwintco/index.php/Hyperspectral_Imagery_Synthesis_tools_for_MATLAB

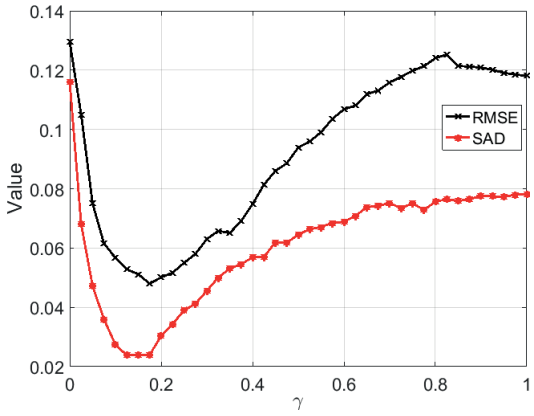


Fig. 7. Performance analysis of SC-GLR$_{l_1}$NMF with respect to γ.

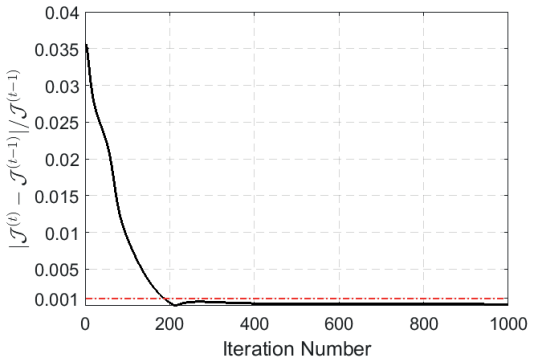


Fig. 8. Convergence analysis of SC-GLR$_{l_1}$NMF.

(SAD) and root-mean-square-error (RMSE) values at each γ are the mean values calculated using unmixing results for all the endmembers. In the experiment, we increased γ from 0 to 1 in 0.025 increments. It can be seen that when $\gamma = 0.175$, both SAD and RMSE values reach their minimums. Thus, in the following experiments, we set γ to 0.175 in order to obtain an accurate HU solution. As described in Section IV-B, the ratio $|\mathcal{J}^{(t)} - \mathcal{J}^{(t-1)}|/\mathcal{J}^{(t-1)}$ defines the stopping criterion. Fig. 8 shows the value of this ratio at each iteration, and it can easily be seen that this ratio drops below 1×10^{-3} around the 200th iteration. Thus, we fix $T_{\max}$ at 300 to ensure the algorithm convergence.

B. Performance Criteria

To quantify the unmixing accuracy, two popular metrics named SAD and RMSE are used. SAD$_k$ measures, in radians, the spectral angle between the estimated endmember signature $\widehat{\mathbf{a}}_k$ and the actual endmember signature $\mathbf{a}_k$ as follows:

$$\text{SAD}_k = \cos^{-1}\left(\frac{\widehat{\mathbf{a}}_k^T \mathbf{a}_k}{\|\widehat{\mathbf{a}}_k\|_2 \|\mathbf{a}_k\|_2}\right).$$

RMSE$_k$ is used to measure the difference between the estimated k^{th} abundance map $\widehat{\mathbf{S}}_k$ and the corresponding ground truth $\mathbf{S}_k$, which is calculated as follows:

$$\text{RMSE}_k = \sqrt{\frac{1}{N}\sum_{n=1}^{N}\left(\mathbf{S}_{ki} - \widehat{\mathbf{S}}_{ki}\right)^2}.$$

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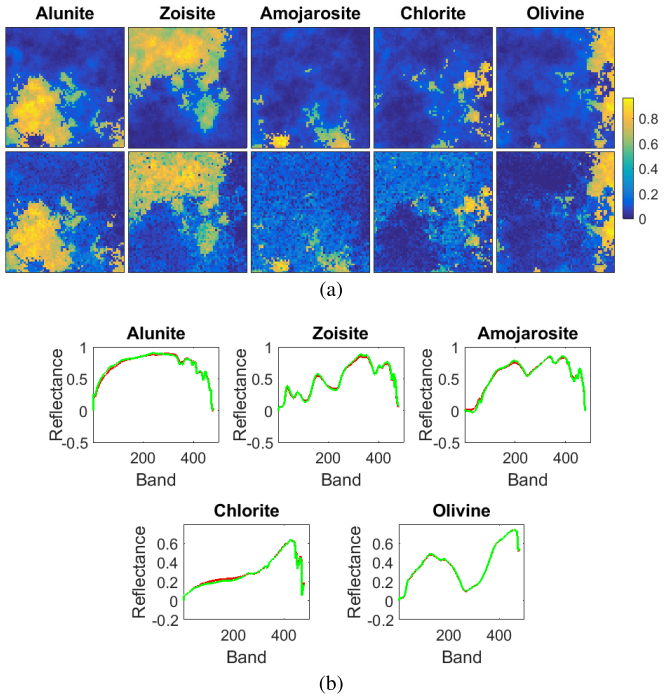


Fig. 9. Abundance and endmember estimation of the simulated data set with SC-GLR$_{l_1}$NMF. (a) (Top row) Ground-truth abundance maps. (Bottom row) Estimated abundance maps. (b) (Red line) Reference endmember signature. (Green line) Estimated endmember signature.

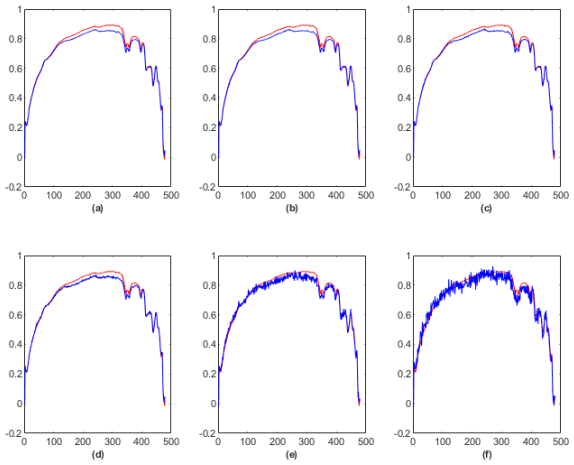


Fig. 10. Endmember estimation of a chosen endmember (in this case "Alunite") from noisy synthetic data at different SNR levels with SC-GLR$_{l_1}$NMF. (Red line) Reference endmember signature. (Blue line) Estimated endmember signature. Noise levels: (a) 50, (b) 40, (c) 30, (d) 20, (e) 10, and (f) 5 dB.

C. Experiments on Synthetic Data

1) Experiment 1 (Unmixing Performance): Here, we compare the unmixing performance of SC-GLR$_{l_1}$NMF with several other HU algorithms. The SAD and RMSE values for each endmember under different methods are tabulated in Tables I and II, respectively. We can easily observe that the proposed method almost achieves the best performance for each case and obtains the smallest SAD average out of the five compared algorithms. Fig. 9(a) and (b), respectively, shows the abundance maps and the endmember signatures estimated via SC-GLR$_{l_1}$NMF with respect to their ground truths.

2) Experiment 2 (Robustness to Noise): The goal of this experiment is to evaluate the robustness of the proposed algorithm to noise. To the original noise-free data set, we add the zero-mean white Gaussian noise with a specific signal-to-noise ratio (SNR)

$$\text{SNR} = 10\log_{10}\frac{E(\mathbf{x}^T\mathbf{x})}{E(\mathbf{n}^T\mathbf{n})}$$

where $\mathbf{x}$ and $\mathbf{n}$, respectively, stand for the uncorrupted spectrum and the noise at a pixel, and $E(\cdot)$ is the expectation operator. The experiment is conducted under six different SNRs: 50, 40, 30, 10, 10, and 5 dB. The estimated endmember signatures of a chosen endmember (for example Alunite in this instance), extracted from the noisy synthetic data using the proposed SC-GLR$_{l_1}$NMF algorithm, at different SNR levels, are depicted in Fig. 10. Also, the mean SAD and RMSE profiles of different algorithms with respect to the noise level are shown in Fig. 11(a) and (b), respectively. In terms of the SAD values, SC-GLR$_{l_1}$NMF has superior robustness properties and remains to achieve the smallest mean SAD values over the range of interest.

When the variation of the mean RMSE values is considered, SGSNMF is slightly more robust to noise than SC-GLR$_{l_1}$NMF. However, the proposed SC-GLR$_{l_1}$NMF is capable of maintaining a smaller mean RMSE than SGSNMF beyond 15 dB. Overall, When SNR is increased beyond 10–15 dB, as shown in Fig. 11(a), SC-GLR$_{l_1}$NMF degrades gradually but still clearly outperforms other existing methods in terms of SAD. In addition, as shown in Fig. 11(b), SC-GLR$_{l_1}$NMF holds the performance at a near-constant level while maintaining a smaller mean RMSE compared with all other methods, providing clear evidence of its comparative robustness to noise. The superior noise rejection capability of SC-GLR$_{l_1}$NMF is mainly due to the fact that GLR$_{l_1}$ aims to extract PWS signals, suppressing rapidly varying noise components.

3) Experiment 3 (Sensitivity to No. of Pixels): In this experiment, we show how the proposed algorithm performs on data sets with different no. of pixels. It is important to assess the performance of the novel algorithm against different sizes of images in order to assess the performance of the algorithm under the availability of different amounts of statistical information. The amount of information influences the algorithm to be or not be trapped on a local minima [3]. Thus, using the HSIST, we create six HSIs of different sizes: 20×20, 40×40, 60×60, 80×80, 100×100, 200×200, and 300×300 pixels. We use the same endmembers in Fig. 6 for the generation of images. Fig. 11(c) and (d), respectively, shows the variation of the mean SAD and RMSE values of different algorithms with respect to the number of pixels. It can be observed from both figures that HG$L_{1/2}$NMF, SGSNMF, and SC-GLR$_{l_1}$NMF algorithms are less sensitive to the number of pixels than the other methods. Furthermore, out of these three, SGSNMF and SC-GLR$_{l_1}$NMF are competitive in the unmixing performance and record the lowest of the mean SAD and RMSE values.

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TABLE I

SAD COMPARISON FOR THE SIMULATED DATA SET. THE BEST RESULTS ARE IN BOLD TYPEFACE, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	SC-GLR$_{l_1}$NMF	SGSNMF	HG$L_{1/2}$-NMF	GLNMF	TV-RSNMF	$l_{1/2}$-NMF
Alunite	<u>0.0261</u>	**0.0222**	0.0349	0.0813	0.0412	0.0834
Zoisite	**0.0263**	<u>0.0515</u>	0.0846	0.1508	0.0708	0.1551
Amojarosite	<u>0.0297</u>	**0.0184**	0.0579	0.0822	0.1215	0.0835
Chlorite	<u>0.0325</u>	**0.0223**	0.1363	0.1186	0.0591	0.0719
Olivine	**0.0045**	0.0856	<u>0.0461</u>	0.0782	0.2900	0.1857
Mean	**0.0238**	<u>0.0400</u>	0.0719	0.1022	0.1165	0.1159

TABLE II

RMSE COMPARISON FOR SIMULATED DATA SET. THE BEST RESULTS ARE IN BOLD TYPEFACE, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	SC-GLR$_{l_1}$NMF	SGSNMF	HG$L_{1/2}$-NMF	GLNMF	TV-RSNMF	$l_{1/2}$-NMF
Alunite	**0.0332**	<u>0.0435</u>	0.0584	0.1806	0.0538	0.1812
Zoisite	**0.0695**	<u>0.0942</u>	0.1617	0.1322	0.1027	0.1327
Amojarosite	<u>0.0499</u>	**0.0360**	0.0904	0.1095	0.0944	0.1393
Chlorite	<u>0.0608</u>	**0.0308**	0.0809	0.1088	0.1149	0.1007
olivine	**0.0266**	<u>0.0310</u>	0.0468	0.1162	0.1223	0.1169
Mean	<u>0.0480</u>	**0.0471**	0.0876	0.1295	0.0976	0.1342

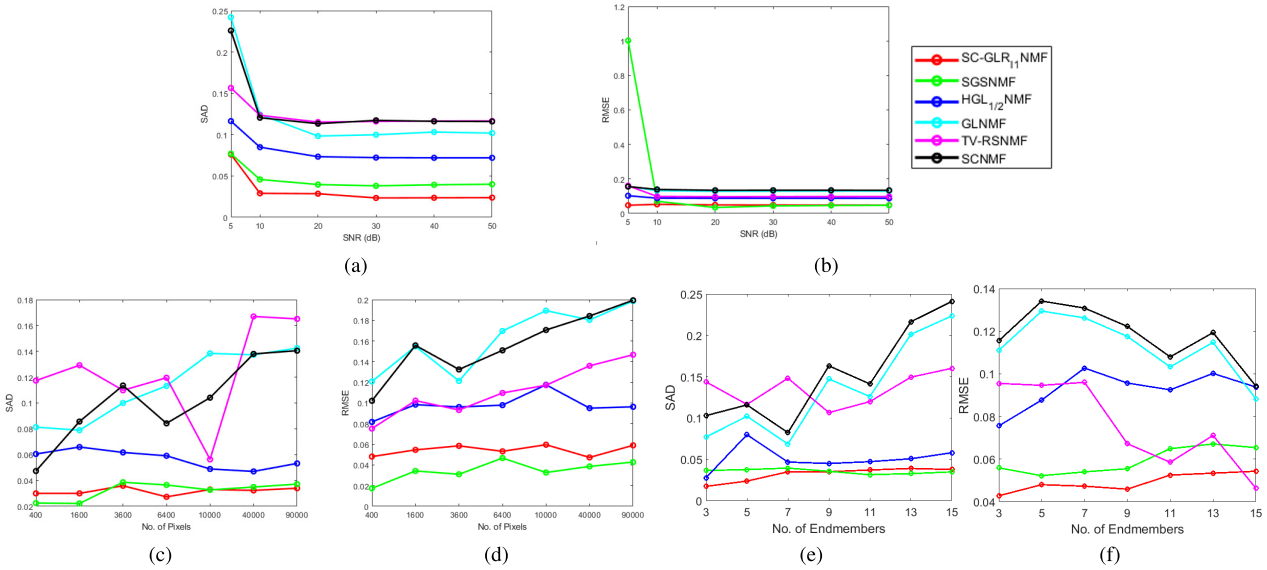


Fig. 11. Results of experiments 3–5. (a) SAD performance with SNR. (b) RMSE performance with SNR. (c) SAD performance with no. of pixels. (d) RMSE performance with no. of pixels. (e) SAD performance with no. of endmembers. (f) RMSE performance with no. of endmembers.

4) Experiment 4 (Sensitivity to No. of Endmembers): We conduct an experiment to assess the influence of the number of constituent endmembers in an HSI scene. For that, we assume that the number of endmembers is known in advance. Using the HSIST, we construct seven 64×64 HSIs having K endmembers, where $K = \{3, 5, 7, 9, 11, 13, 15\}$. Fig. 11(e) and (f), respectively, shows the mean SAD and RMSE profiles of different algorithms with respect to the number of endmembers. It can be observed from these figures that both SGSNMF and SC-GLR$_{l_1}$NMF are much more resistant to the variation of the number of endmembers than the rest. The SAD performances of both SGSNMF and SC-

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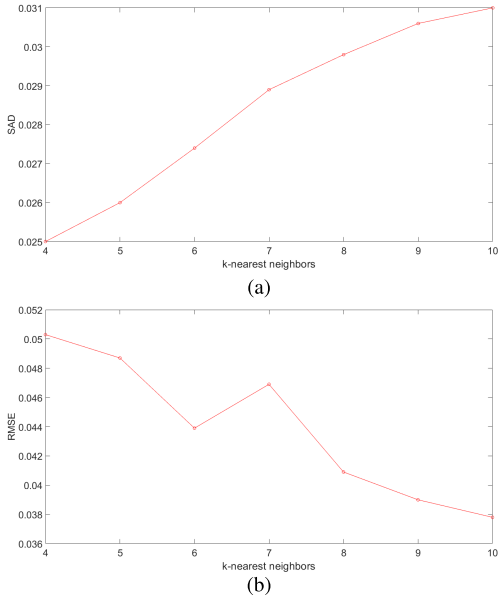


Fig. 12. Results of experiment 5. Variation of (a) mean SAD and (b) mean RMSE of the proposed algorithm against the number of nearest neighbors considered when incorporating the k-nearest neighbor graph construction.

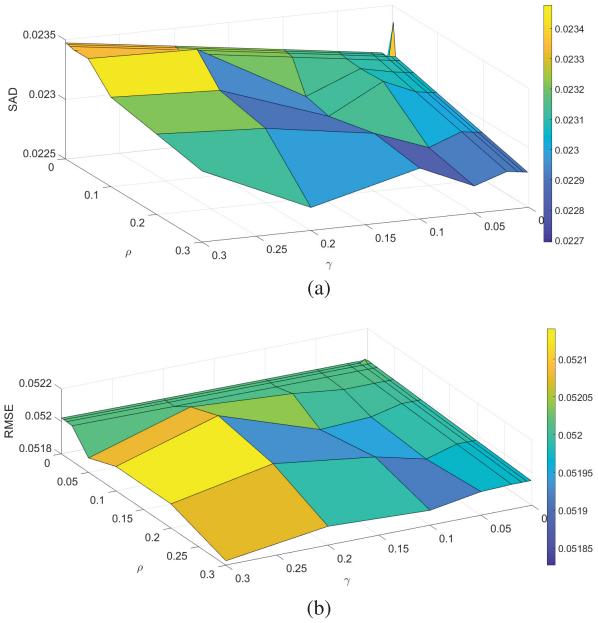


Fig. 13. Results of experiment 6. Variation of (a) mean SAD and (b) mean RMSE of the proposed algorithm against the positive parameters γ and ρ in Problem 4.

GLR$_{l_1}$NMF are almost the same and are plainly superior to those of others. In terms of the RMSE performance, SC-GLR$_{l_1}$NMF is better than SGSNMF. As the number of endmembers increases, TV-RSNMF exhibits a downward trend in its RMSE profile. Therefore, its RMSE performance even surpasses that of SC-GLR$_{l_1}$NMF when a large number of endmembers are present in the HSI scene. However, TV-RSNMF is much more sensitive to the number of endmembers than SC-GLR$_{l_1}$NMF. Thus, in situations when an HU algorithm requires to process HSIs having a different number of endmembers, then SC-GLR$_{l_1}$NMF would be the most suitable choice.

5) Experiment 5 (Sensitivity to the Graph Structure): In this experiment, we show how the proposed algorithm performs when changing the parameters in Problem 4. In the previous experiments, we have utilized a four-connectivity graph structure. Thus, to assess the performance of the proposed graph-based unmixing method's dependence on graph construction, we conduct an experiment utilizing another popular graph structure, i.e., the k-nearest neighbor structure. In Fig. 12, we present the results for the k-nearest neighbor graphs, where $k = \{4, 5, 6, 7, 8, 9, 10\}$. The HSI data set used here is the same simulated data set with five endmembers. Fig. 12(a) and (b), respectively, shows the mean SAD and RMSE profiles of the proposed algorithm with respect to the considered k-nearest neighbor-based graph structures.

6) Experiment 6 (Sensitivity to Control Parameters): In this experiment, we show how the proposed algorithm performs when varying the control parameters γ and ρ in Problem 4. Fig. 13(a) and (b), respectively, shows the mean SAD and mean RMSE as 3-D surface plots against the variation of γ and ρ. Even though the performance variation does not depict a clear trend, this experiment would be useful in selecting the desired parameter values to obtain the optimum performance.

D. Evaluation With Real Hyperspectral Data

In order to conduct real-data experiments, we chose three real-world data sets: the Jasper Ridge data set, the Urban Hyperspectral Digital Imagery Collection Experiment (HYDICE) data set, and the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) Cuprite data set. The parameters used for SC-GLR$_{l_1}$NMF algorithm in the following experiments are as follows: $\sigma = 1$, $\gamma = 0.175$, $T_{\max} = 300$, and ρ is calculated for each data set using (27).

The ground-truth abundance maps for all real data sets had been generated using the procedure described by proposed by Jia and Qian [60], [61]. Ground truths generated in a similar fashion had been utilized by many researchers in the recent past for the comparison of various unmixing algorithms [24], [62]. In this method, first, the virtual dimensionality algorithm proposed by Chang and Du [57] is utilized to determine the number of endmembers in the HSI. Second, the pixels containing the pure endmember spectra are manually chosen. These pure spectra should satisfy a high correlation with the reference hyperspectral spectra that are available in the USGS mineral spectral library. Finally, based on the chosen endmember spectra, the corresponding abundances are determined by solving a constrained convex optimization problem, using the CVX Optimization Toolbox in MATLAB. The ground-truth abundance maps generated according to the above-mentioned method can be found in [63]. These readily available ground-truth data have been utilized in this article in order to assess and compare the performance of the proposed SC-GLR$_{l_1}$NMF method.

1) Jasper Data Set: The Jasper Ridge data set is a 512 × 614 image, collected by the airborne imaging spectrometer sensor over Jasper Ridge in central California, USA. The HSI contains 224 bands covering the electromagnetic spectrum ranging from 0.38 to 2.5 μm. For the experiments, a subimage

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TABLE III
SAD COMPARISON FOR THE JASPER DATA SET. THE BEST RESULTS ARE IN BOLD TYPEFACE, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	SC-GLR$_{l_1}$NMF	SGSNMF	HG$L_{1/2}$-NMF	GLNMF	TV-RSNMF	$l_{1/2}$-NMF
Tree	**0.0100**	<u>0.0108</u>	0.1854	0.1120	0.0180	0.1369
Water	0.0503	<u>0.0501</u>	0.1251	**0.0308**	0.0822	0.1150
Soil	<u>0.0356</u>	**0.0052**	0.0790	0.1128	0.0760	0.1026
Road	**0.0171**	0.2268	<u>0.1034</u>	0.1259	0.1519	0.1989
Mean	**0.0283**	<u>0.0732</u>	0.1232	0.0954	0.0820	0.1384

TABLE IV
RMSE COMPARISON FOR THE JASPER DATA SET. THE BEST RESULTS ARE IN BOLD TYPEFACE, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	SC-GLR$_{l_1}$NMF	SGSNMF	HG$L_{1/2}$-NMF	GLNMF	TV-RSNMF	$l_{1/2}$-NMF
Tree	<u>0.0261</u>	**0.0069**	0.1711	0.0708	0.0875	0.2348
Water	<u>0.0215</u>	**0.0070**	0.0915	0.0438	0.0450	0.1890
Soil	**0.0735**	0.1125	0.0969	0.1051	0.1009	<u>0.0934</u>
Road	**0.0582**	0.2736	<u>0.1225</u>	0.2424	0.2178	0.1861
Mean	**0.0448**	<u>0.1000</u>	0.1205	0.1155	0.1128	0.1758

with 100 × 100 pixels and 198 bands is used after removing low-SNR and water-vapor absorption bands. There are four distinct target endmembers in the HSI scene: "1# Tree," "2# Water," "3# Soil," and "4# Road."

The SAD and RMSE values for each endmember under different algorithms are tabulated in Tables III and IV, respectively. The endmemberwise unmixing results of SC-GLR$_{l_1}$NMF are competitive with that of SGSNMF, but the former achieves the lowest mean SAD and RMSE values out of all the algorithms. The abundance maps and the endmember signatures estimated by SC-GLR$_{l_1}$NMF are shown in the bottom row of Fig. 14(a) and (b), respectively, and it is apparent that they are in accordance with their respective ground truths. The estimated endmember signature of "3# Tree" almost coincides with the true signature. These reveal that the proposed algorithm outperforms the other popular algorithms used in the comparison.

2) Urban HYDICE Data Set: The original Urban data set is a 307 × 307 pixel image having 210 bands with spectral coverage from 0.4 to 2.5 μm. We remove some low-SNR bands and the water-vapor bands (1–4, 76, 87, 101—111, 136—153, and 198—210) and retain the remaining 162 bands. There are four distinct target endmembers in the HSI scene: "1# Asphalt," "2# Grass," "3# Tree," and "4# Roof."

The comparison of SAD and RMSE values of different algorithms is presented in Tables V and VI, respectively. In terms of the SAD performance, SGSNMF outperforms all the other algorithms; however, the proposed algorithm SC-GLR$_{l_1}$NMF closely follows SGSNMF, and its mean SAD value is almost on par with that of SGSNMF. The RMSE performance of SC-GLR$_{l_1}$NMF, on the other hand, is clearly better than that of the others. The abundance maps and the endmember signatures estimated by SC-GLR$_{l_1}$NMF are shown in the bottom row of Fig. 15(a) and (b), respectively, with respect to their ground truths.

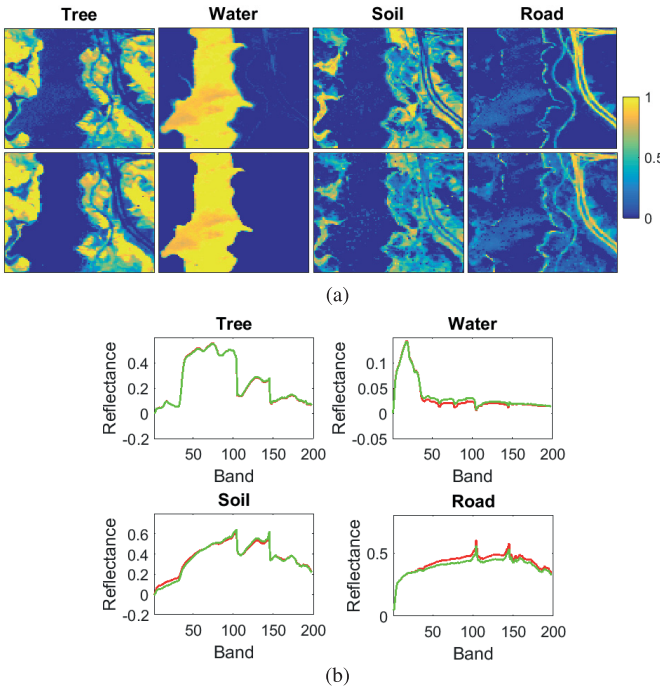


Fig. 14. Abundance and endmember estimation of the Jasper data set with SC-GLR11NMF. (a) (Top row) Ground-truth abundance maps. (Bottom row) Estimated abundance maps. (b) (Red line) Reference endmember signature. (Green line) Estimated endmember signature.

3) Cuprite Data Set: The AVIRIS Cuprite data contain 250 × 191 pixels, and each pixel has 224 bands ranging from 0.4 to 2.5 μm. As tabulated in Table VII, there are mainly

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TABLE V

SAD COMPARISON FOR URBAN DATA SET. THE BEST RESULTS ARE IN BOLD TYPEFACE, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	SC-GLR$_{l_1}$NMF	SGSNMF	HG$L_{1/2}$-NMF	GLNMF	TV-RSNMF	$l_{1/2}$-NMF
Asphalt	<u>0.0235</u>	**0.0151**	0.0817	0.1074	0.1016	0.1014
Grass	<u>0.0164</u>	**0.0175**	0.0524	0.0757	0.1145	0.1233
Tree	<u>0.0240</u>	**0.0072**	0.0734	0.1115	0.1084	0.0506
Roof	**0.0117**	<u>0.0122</u>	0.0842	0.1382	0.1066	0.1543
Mean	<u>0.0189</u>	**0.0130**	0.0729	0.1082	0.1078	0.1074

TABLE VI

RMSE COMPARISON FOR URBAN DATA SET. THE BEST RESULTS ARE IN BOLD TYPEFACE, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	SC-GLR$_{l_1}$NMF	SGSNMF	HG$L_{1/2}$-NMF	GLNMF	TV-RSNMF	$l_{1/2}$-NMF
Asphalt	**0.0202**	<u>0.0284</u>	0.0678	0.1274	0.1105	0.1016
Grass	**0.0238**	<u>0.0363</u>	0.0917	0.1115	0.1306	0.1069
Tree	**0.0137**	<u>0.0358</u>	0.0508	0.0257	0.0578	0.1303
Roof	<u>0.0159</u>	**0.0117**	0.1267	0.1808	0.1561	0.1597
Mean	**0.0184**	<u>0.0281</u>	0.0843	0.1113	0.1138	0.1246

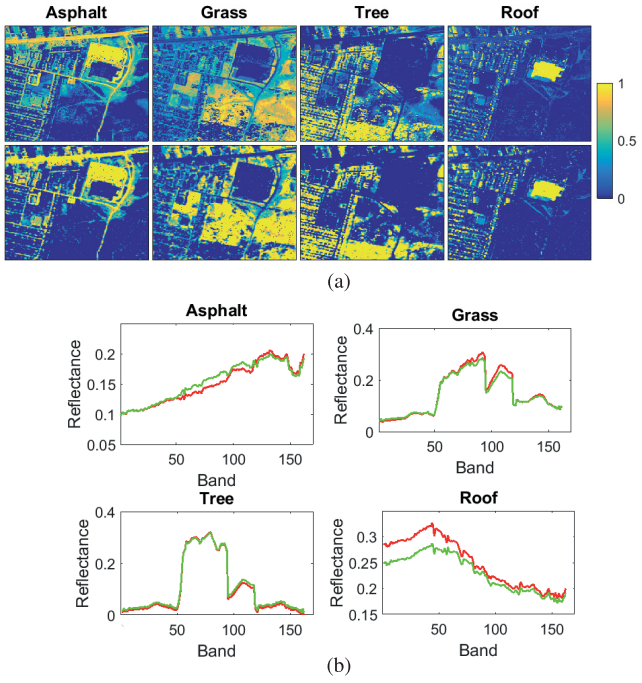


Fig. 15. Abundance and endmember estimation of the Urban data set with SC-GLRl1NMF. (a) (Top row) Ground-truth abundance maps. (Bottom row) Estimated abundance maps. (b) (Red line) Reference endmember signature. (Green line) Estimated endmember signature.

12 endmembers in the HSI scene. Only the SAD performances of the algorithms are presented as the groud truths for the abundance maps are not available. According to Table VII, SC-GLR$_{l_1}$NMF achieves the overall best performance (the smallest mean SAD value), while SGSNMF, HG$L_{1/2}$-NMF, and TV-RSNMF demonstrate competitive endmemberwise

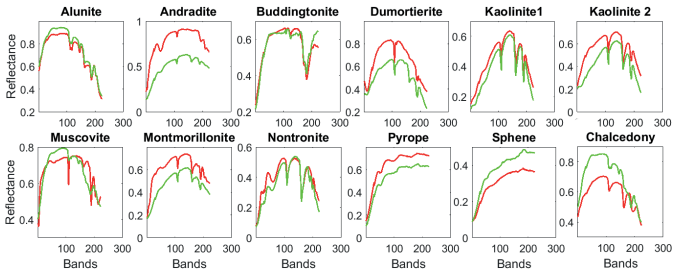


Fig. 16. Endmember estimation of the Cuprite data set with SC-GLRl1NMF. (Red line) Reference endmember signature. (Green line) Estimated endmember signature.

unmixing performance with the proposed algorithm. The endmember signatures estimated by SC-GLR$_{l_1}$NMF are shown in Fig. 16, and it can be observed that they closely follow their reference signatures.

Apart from the SAD performance comparison, we present a run-time evaluation with all the other competing methods on the AVIRIS Cuprite data. All experiments are carried out on a Windows system with a 2.6-GHz Intel Core i5-3230 processor using MATLAB 2015a. The results are summarized in Table VIII. In terms of the average run time, $l_{1/2}$-NMF outperforms all the other algorithms; however, it can be seen that the proposed algorithm SC-GLR$_{l_1}$NMF has a comparable time consumption with that of HG$L_{1/2}$-NMF and GLNMF while highly outperforming SGSNMF. As stated earlier, it is apparent that the proposed SC-GLR$_{l_1}$NMF achieves the best performance in terms of SAD compared with all the other methods considered. The $l_{1/2}$-NMF algorithm that has a much shorter run time compared with the rest of the algorithm clearly has poor accuracy performance as it is outperformed by most algorithms in terms of accuracy for all considered

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TABLE VII
SAD COMPARISON FOR CUPRITE DATA SET. THE BEST RESULTS ARE IN BOLD TYPEFACE, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	SC-GLR$_{l_1}$NMF	SGSNMF	HG$L_{1/2}$-NMF	GLNMF	TV-RSNMF	$l_{1/2}$-NMF
Alunite	**0.0165**	0.0346	0.0425	0.0419	0.1734	0.0766
Andradite	0.0850	0.1024	0.0645	0.0886	**0.0582**	0.0971
Buddingtonite	0.0544	**0.0282**	0.0467	0.1771	0.0935	0.0885
Dumortierite	**0.0567**	0.1542	0.0960	0.1250	0.0881	0.0753
Kaolinite1	0.0784	**0.0097**	0.1118	0.4160	0.0593	0.1747
Kaolinite2	0.1201	0.1211	0.0611	0.0948	**0.0476**	0.0785
Muscovite	**0.0801**	0.1837	0.0979	0.1579	0.0969	0.1258
Montmorillonite	0.0902	0.1075	0.0976	0.0725	**0.0514**	0.0594
Nontronite	0.0918	0.1273	0.1241	**0.0878**	0.1111	0.0892
Pyrope	0.0349	**0.0070**	0.0635	0.0663	0.0983	0.1057
Sphene	0.0702	**0.0301**	0.1662	0.1104	0.1447	0.1506
Chalcedony	0.0773	**0.0660**	0.1092	0.1251	0.0960	0.1063
Mean	**0.0713**	0.0809	0.0901	0.1303	0.0932	0.1023

TABLE VIII
AVERAGE RUN-TIME (IN SECONDS) COMPARISON FOR THE CUPRITE DATA SET. THE BEST RESULTS ARE IN BOLD TYPEFACE, AND THE SECOND-BEST RESULTS ARE UNDERLINED

Methods	SC-GLR$_{l_1}$NMF	SGSNMF	HG$L_{1/2}$-NMF	GLNMF	TV-RSNMF	$l_{1/2}$-NMF
Run-time	55.25	80.83	52.30	45.47	32.40	**18.59**

databases. The rest of the algorithms have comparable run time to the proposed algorithm, with the exception of the SGSNMF algorithm that suffers severely in terms of run time. Therefore, it can be concluded that the proposed method while having comparable run time to most of the existing state-of-the-art techniques clearly outperforms them in terms of HU accuracy.

VI. CONCLUSION

This article proposed a novel blind HU algorithm called SC-GLR$_{l_1}$NMF. The proposed algorithm utilizes the $l_{1/2}$-sparsity constraint coupled with the newly developed GLR$_{l_1}$ regularization framework to enrich the NMF, required for the HU problem. The motivation for regularizing with GLR$_{l_1}$ was to extract PWS abundance maps from observed HSIs. Through a GSP perspective, it was shown that GLR$_{l_1}$ has natural PWS signal generation and noise suppression capabilities. These attributes have enabled the proposed SC-GLR$_{l_1}$NMF to extract endmember signatures and abundance maps with great precision for experiments conducted on both synthetic and real data sets. Moreover, experimental results verify that it can yield state-of-the-art unmixing performance through the introduction of the proposed smoothness constraint.

APPENDIX A
SOLUTION TO PROBLEM 3

Let the objective function associated with Problem 3 be defined as

$$\mathbb{V} = \|\mathbf{X} - \mathbf{AS}\|_F^2 + \gamma \operatorname{tr}(\mathbf{S}\mathcal{L}\mathbf{S}^T). \tag{28}$$

Then, we can show that

$$\mathbb{V} = \operatorname{tr}\left(\mathbf{X}^T\mathbf{X}\right) - 2\operatorname{tr}\left(\mathbf{X}^T\mathbf{AS}\right) + \operatorname{tr}\left(\mathbf{S}^T\Delta\mathbf{S}\right) + \gamma \operatorname{tr}(\mathbf{S}\mathcal{L}\mathbf{S}^T)$$

where $\Delta = \mathbf{A}^T\mathbf{A}$. As shown in Appendix B, if it is assumed that the matrix $\mathbf{S}$ is not so sparse and K is large, then $\partial\left(\operatorname{tr}(\mathbf{S}\mathcal{L}\mathbf{S}^T)\right)/\partial\mathbf{S}_{kn} = 2(\mathbf{S}\mathcal{L})_{kn}$. Therefore, under this assumption, the first derivative of $\mathbb{V}$ with respect to $\mathbf{S}_{kn}$ can be written as

$$\frac{\partial\mathbb{V}}{\partial\mathbf{S}_{kn}} = -2(\mathbf{A}^T\mathbf{X})_{kn} + 2(\Delta\mathbf{S})_{kn} + 2\gamma\,(\mathbf{S}\mathcal{L})_{kn}$$

noting that $\Delta^T = \Delta$. Its second derivative is obtained as

$$\frac{\partial^2\mathbb{V}}{\partial\mathbf{S}_{kn}^2} = 2\Delta_{kk} + 2\gamma\,\mathcal{L}_{nn}.$$

Since both Δ_{kk} and $\mathcal{L}_{nn}$ are positive, $\mathbb{V}$ is convex in all $\mathbf{S}_{kn}$. Therefore, $\mathbb{V}$ has a globally minimum at $\mathbf{S}$, which satisfies

$$\begin{aligned} -2\mathbf{A}^T\mathbf{X} + 2\Delta\mathbf{S} + 2\gamma\,\mathbf{S}\mathcal{L} &= 0 \\ \implies \Delta\mathbf{S} + \gamma\,\mathbf{S}\mathcal{L} &= \mathbf{A}^T\mathbf{X}. \end{aligned} \tag{29}$$

This is a Sylvester equation in $\mathbf{S}$. As both Δ and $\mathcal{L}$ are real and symmetric, we can denote their eigendecompositions as $\mathbf{U}^T\Delta\mathbf{U} = \operatorname{diag}(\mu_1, \ldots, \mu_K)$ and $\mathbf{V}^T\mathcal{L}\mathbf{V} = \operatorname{diag}(\lambda_1, \ldots, \lambda_N)$. Then, a solution to (29) can be obtained as [64]

$$\mathbf{S} = \mathbf{U}\widetilde{\mathbf{S}}\mathbf{V}^T \tag{30}$$

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where

$$\widetilde{\mathbf{S}}_{ij} = \frac{\mathbf{C}_{ij}}{\mu_i + \gamma\,\lambda_j}, \text{ for } \mathbf{C} = \mathbf{U}^T(\mathbf{A}^T\mathbf{X})\mathbf{V}.$$

This solution exists if and only if $\mu_i + \gamma\,\lambda_j \neq 0$, i.e., when Δ and $-\gamma\,\mathcal{L}$ do not have common eigenvalues. Then, recalling that $\mathbf{V} = [\mathbf{v}_1, \ldots, \mathbf{v}_N]$ and letting $\mathbf{Q} = \mathbf{U}\widetilde{\mathbf{S}}$, we have $\mathbf{S} = \mathbf{Q}\mathbf{V}^T$. If we consider $\mathbf{S}_m$ which is, in fact, the mth graph signal, we have $\mathbf{S}_m = \sum_{n=1}^{N} \mathbf{Q}_{mn}\mathbf{v}_n^T$. The mnth element of $\mathbf{Q}$ can be written as

$$\mathbf{Q}_{mn} = \sum_{k=1}^{K} \mathbf{U}_{mk}\widetilde{\mathbf{S}}_{kn} = \sum_{k=1}^{K} \mathbf{U}_{mk}\frac{\mathbf{C}_{kn}}{\mu_k + \gamma\,\lambda_n}.$$

Thus, we can write that

$$\mathbf{S}_m = \sum_{n=1}^{N}\left(\sum_{k=1}^{K} \mathbf{U}_{mk}\mathbf{C}_{kn}\frac{\mathbf{v}_n^T}{\mu_k + \gamma\,\lambda_n}\right). \tag{31}$$

This solution is exact when the Laplacian matrix $\mathbf{L}$ is used instead of $\mathcal{L}$ as $\partial\big(\mathrm{tr}\,(\mathbf{S}\mathbf{L}\mathbf{S}^T)\big)/\partial\mathbf{S}_{kn} = 2(\mathbf{S}\mathbf{L})_{kn}$ under no assumption.

Appendix B
Derivation of (23) and (24)

For the ease of presentation, we denote the partial derivative of a scalar x with respect to $\mathbf{S}_{kn}$ as $\partial_{kn}(x)$, i.e., $\partial x/\partial(\mathbf{S}_{kn}) = \partial_{kn}(x)$. Let us first consider F, which is

$$F = \mathrm{tr}(\mathbf{S}\mathcal{W}\mathbf{S}^T) = \sum_{i,j,m} \mathbf{S}_{ij}\mathcal{W}_{jm}\mathbf{S}_{im}.$$

We can find its first derivative with respect to $\mathbf{S}_{kn}$ as

$$\begin{aligned}\partial_{kn}(F) &= \sum_{i,j,m} \partial_{kn}(\mathbf{S}_{ij})\mathcal{W}_{jm}\mathbf{S}_{im} + \sum_{i,j,m} \mathbf{S}_{ij}\mathcal{W}_{jm}\partial_{kn}(\mathbf{S}_{im}) \\ &\quad + \sum_{i,j,m} \mathbf{S}_{ij}\partial_{kn}(\mathcal{W}_{jm})\mathbf{S}_{im} \\ &= \sum_{m} \mathcal{W}_{nm}\mathbf{S}_{km} + \sum_{j} \mathbf{S}_{kj}\mathcal{W}_{jn} \\ &\quad + \sum_{i,j,m} \mathbf{S}_{ij}\partial_{kn}(\mathcal{W}_{jm})\mathbf{S}_{im} \\ &= 2(\mathbf{S}\mathcal{W})_{kn} + \sum_{i,j,m} \mathbf{S}_{ij}\partial_{kn}(\mathcal{W}_{jm})\mathbf{S}_{im} \end{aligned} \tag{32}$$

recalling the symmetric property of $\mathcal{W}$. We can write from (18) that

$$\begin{aligned}&\sum_{i,j,m} \mathbf{S}_{ij}\partial_{kn}(\mathcal{W}_{jm})\mathbf{S}_{im} \\ &\quad= \sum_{\substack{i,j,m \text{ s.t.} \\ (m,j)\in\mathcal{E}}} \mathbf{S}_{ij}\mathbf{W}_{jm}\partial_{kn}\left(\frac{\|\mathbf{s}_j - \mathbf{s}_m\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_j - \mathbf{s}_m\|_2^2 + \epsilon^2}\right)\mathbf{S}_{im} \\ &\quad= \sum_{\substack{i,m \text{ s.t.} \\ (m,n)\in\mathcal{E}}} \mathbf{S}_{in}\mathbf{W}_{nm}\partial_{kn}\left(\frac{\|\mathbf{s}_n - \mathbf{s}_m\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2}\right)\mathbf{S}_{im} \\ &\qquad+ \sum_{\substack{i,j \text{ s.t.} \\ (j,n)\in\mathcal{E}}} \mathbf{S}_{ij}\mathbf{W}_{jn}\partial_{kn}\left(\frac{\|\mathbf{s}_j - \mathbf{s}_n\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_j - \mathbf{s}_n\|_2^2 + \epsilon^2}\right)\mathbf{S}_{in}. \end{aligned} \tag{33}$$

For all m, n subject to $(m, n) \in \mathcal{E}$, we can show that

$$\begin{aligned}&\partial_{kn}\left(\frac{\|\mathbf{s}_n - \mathbf{s}_m\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2}\right) \\ &= \frac{\sum_r \mathbf{z}_{nm}(r)\big(\partial_{kn}(\mathbf{S}_{rn} - \mathbf{S}_{rm})\big)}{\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2} \\ &\quad - 2\frac{(\|\mathbf{s}_n - \mathbf{s}_m\|_1 + \sqrt{K}\epsilon)\sum_l(\mathbf{S}_{ln} - \mathbf{S}_{lm})\big(\partial_{kn}(\mathbf{S}_{ln} - \mathbf{S}_{lm})\big)}{\big(\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2\big)^2} \\ &= \frac{\mathbf{z}_{nm}(k)}{\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon} - 2\frac{(\mathbf{S}_{kn} - \mathbf{S}_{km})(\|\mathbf{s}_n - \mathbf{s}_m\|_1 + \sqrt{K}\epsilon)}{\big(\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2\big)^2} \end{aligned} \tag{34}$$

where $\mathbf{z}_{nm}(k)$ is defined in (14).

In a similar manner, for all j, n subject to $(j, n) \in \mathcal{E}$, we can show that

$$\begin{aligned}&\partial_{kn}\left(\frac{\|\mathbf{s}_j - \mathbf{s}_n\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_j - \mathbf{s}_n\|_2^2 + \epsilon^2}\right) \\ &= -\frac{\mathbf{z}_{jn}(k)}{\|\mathbf{s}_j - \mathbf{s}_n\|_2^2 + \epsilon} + 2\frac{(\mathbf{S}_{kj} - \mathbf{S}_{kn})(\|\mathbf{s}_j - \mathbf{s}_n\|_1 + \sqrt{K}\epsilon)}{\big(\|\mathbf{s}_j - \mathbf{s}_n\|_2^2 + \epsilon^2\big)^2}. \end{aligned} \tag{35}$$

As we assume that the abundances on neighboring pixels are similar, we have that $\mathbf{S}_{kn} - \mathbf{S}_{km} \approx 0$ in (34) and $\mathbf{S}_{kj} - \mathbf{S}_{kn} \approx 0$ in (35). Then, it follows from (33) that

$$\begin{aligned}&\sum_{i,j,m} \mathbf{S}_{ij}\partial_{kn}(\mathcal{W}_{jm})\mathbf{S}_{im} \\ &\quad= \sum_{\substack{i,m \text{ s.t.} \\ (m,n)\in\mathcal{E}}} \mathbf{S}_{in}\mathbf{W}_{nm}\frac{\mathbf{z}_{nm}(k)}{\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2}\mathbf{S}_{im} \\ &\qquad- \sum_{\substack{i,j \text{ s.t.} \\ (j,n)\in\mathcal{E}}} \mathbf{S}_{ij}\mathbf{W}_{jn}\frac{\mathbf{z}_{jn}(k)}{\|\mathbf{s}_j - \mathbf{s}_n\|_2^2 + \epsilon^2}\mathbf{S}_{in} \\ &\quad= \sum_{\substack{m \text{ s.t.} \\ (m,n)\in\mathcal{E}}} \frac{\mathbf{z}_{nm}(k)\mathbf{W}_{nm}(\mathbf{S}^T\mathbf{S})_{nm}}{\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2} - \sum_{\substack{j \text{ s.t.} \\ (j,n)\in\mathcal{E}}} \frac{\mathbf{z}_{jn}(k)\mathbf{W}_{jn}(\mathbf{S}^T\mathbf{S})_{jn}}{\|\mathbf{s}_j - \mathbf{s}_n\|_2^2 + \epsilon^2} \\ &\quad= \sum_{\substack{m \text{ s.t.} \\ (m,n)\in\mathcal{E}}} \frac{\big(\mathbf{z}_{nm}(k) - \mathbf{z}_{mn}(k)\big)\mathbf{W}_{nm}}{\|\mathbf{s}_n - \mathbf{s}_m\|_2^2 + \epsilon^2}(\mathbf{S}^T\mathbf{S})_{nm}. \end{aligned} \tag{36}$$

The last step of (36) is obtained by change of variables and the symmetric properties of $\mathcal{W}$ and $\mathbf{S}^T\mathbf{S}$.

Then, let us consider G, which is

$$G = \mathrm{tr}(\mathbf{S}\mathcal{D}\mathbf{S}^T) = \sum_{i,j,m} \mathbf{S}_{ij}\mathcal{D}_{jm}\mathbf{S}_{im}.$$

We can find its first derivative with respect to $\mathbf{S}_{kn}$ as

$$\begin{aligned}\partial_{kn}(G) &= \sum_{i,j,m} \partial_{kn}(\mathbf{S}_{ij})\mathcal{D}_{jm}\mathbf{S}_{im} + \sum_{i,j,m} \mathbf{S}_{ij}\mathcal{D}_{jm}\partial_{kn}(\mathbf{S}_{im}) \\ &\quad + \sum_{i,j,m} \mathbf{S}_{ij}\partial_{kn}(\mathcal{D}_{jm})\mathbf{S}_{im} \\ &= 2(\mathbf{S}\mathcal{D})_{kn} + \sum_{ij} \mathbf{S}_{ij}^2\partial_{kn}(\mathcal{D}_{jj}) \end{aligned} \tag{37}$$

using the fact that the off-diagonal elements of $\mathcal{D}$ are zero. Recalling that $\mathcal{D}_{jj} = \sum_p \mathcal{W}_{jp}$ and assuming abundances on neighboring

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pixels are similar, we can deduce using (35) and (36) that

$$\begin{aligned}
&\sum_{ij} \mathbf{S}_{ij}^2 \partial_{kn}(\boldsymbol{\mathcal{D}}_{jj}) \\
&\quad = \sum_{i,j,p} \mathbf{S}_{ij}^2 \mathbf{W}_{jp} \partial_{kn} \left(\frac{\|\mathbf{s}_j - \mathbf{s}_p\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_j - \mathbf{s}_p\|_2^2 + \epsilon^2} \right) \\
&\quad = \sum_{\substack{i,p \text{ s.t.} \\ (p,n)\in\mathcal{E}}} \mathbf{S}_{in}^2 \mathbf{W}_{np} \partial_{kn} \left(\frac{\|\mathbf{s}_n - \mathbf{s}_p\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_n - \mathbf{s}_p\|_2^2 + \epsilon^2} \right) \\
&\qquad + \sum_{\substack{i,j \text{ s.t.} \\ (j,n)\in\mathcal{E}}} \mathbf{S}_{ij}^2 \mathbf{W}_{jn} \partial_{kn} \left(\frac{\|\mathbf{s}_j - \mathbf{s}_n\|_1 + \sqrt{K}\epsilon}{\|\mathbf{s}_j - \mathbf{s}_n\|_2^2 + \epsilon^2} \right) \\
&\quad = \sum_{\substack{i,p \text{ s.t.} \\ (p,n)\in\mathcal{E}}} \left(\mathbf{z}_{np}(k)\mathbf{S}_{in}^2 - \mathbf{z}_{pn}(k)\mathbf{S}_{ip}^2 \right) \frac{\mathbf{W}_{np}}{\|\mathbf{s}_n - \mathbf{s}_p\|_2^2 + \epsilon^2}.
\end{aligned}$$

Remark 2: Note that when the abundance matrix is not so sparse and K is large, then it is unlikely that $\boldsymbol{\mathcal{W}}_{ij}$ given by (18) is going to be dependent on a particular abundance value. Therefore, in such a situation, both $\partial_{kn}(\boldsymbol{\mathcal{W}}_{jm})$ in (32) and $\partial_{kn}(\boldsymbol{\mathcal{D}}_{jj})$ in (37) are almost negligible. In addition, when K is large and the matrix is not sparse, $\mathbf{S}_{ij} \ll 1$. As a result, the rightmost terms in (32) and (37) are negligible with respect to the other terms, leading to the simplified form $\partial_{kn}\left(\operatorname{tr}(\mathbf{S}\boldsymbol{\mathcal{L}}\mathbf{S}^T\right) = \partial_{kn}(G) - \partial_{kn}(F) = 2\big(\mathbf{S}(\boldsymbol{\mathcal{D}} - \boldsymbol{\mathcal{W}})\big)_{kn} = 2(\mathbf{S}\boldsymbol{\mathcal{L}})_{kn}$.

REFERENCES

[1] X.-R. Feng, H.-C. Li, J. Li, Q. Du, A. Plaza, and W. J. Emery, "Hyperspectral unmixing using sparsity-constrained deep nonnegative matrix factorization with total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 6245–6257, Oct. 2018.

[2] J. M. Bioucas-Dias, A. Plaza, G. Camps-Valls, P. Scheunders, N. Nasrabadi, and J. Chanussot, "Hyperspectral remote sensing data analysis and future challenges," *IEEE Geosci. Remote Sens. Mag.*, vol. 1, no. 2, pp. 6–36, Jun. 2013.

[3] W. Wang, Y. Qian, and Y. Y. Tang, "Hypergraph-regularized sparse NMF for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 2, pp. 681–694, Feb. 2016.

[4] E. Ekanayake *et al.*, "Mapping ilmenite deposit in Pulmudai, Sri Lanka using a hyperspectral imaging-based surface mineral mapping method," *J. Nat. Sci. Found. Sri Lanka*, vol. 47, no. 3, p. 271, Sep. 2019, doi: 10.4038/jnsfsr.v47i3.9276.

[5] S. S. P. Vithana *et al.*, "Adaptive hierarchical clustering for hyperspectral image classification: Umbrella clustering," *J. Spectral Imag.*, vol. 8, no. 1, p. a11, Jul. 2019, doi: 10.1255/jsi.2019.a11.

[6] E. M. M. B. Ekanayake *et al.*, "A semi-supervised algorithm to map major vegetation zones using satellite hyperspectral data," in *Proc. 9th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Sep. 2018, pp. 1–5.

[7] D. Ratnayake *et al.*, "Engineering stress in thin films for the field of bistable MEMS," *J. Micromech. Microeng.*, vol. 25, no. 12, Nov. 2015, Art. no. 125025, doi: 10.1088/0960-1317/25/12/125025.

[8] F. Xiong, Y. Qian, J. Zhou, and Y. Y. Tang, "Hyperspectral unmixing via total variation regularized nonnegative tensor factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2341–2357, Apr. 2019.

[9] Y. Qu *et al.*, "Hyperspectral anomaly detection through spectral unmixing and dictionary-based low-rank decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4391–4405, Aug. 2018.

[10] H. Zhang, H. Zhai, L. Zhang, and P. Li, "Spectral–spatial sparse subspace clustering for hyperspectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3672–3684, Jun. 2016.

[11] Q. Wang, X. He, and X. Li, "Locality and structure regularized low rank representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 911–923, Feb. 2019.

[12] W. He, H. Zhang, and L. Zhang, "Total variation regularized reweighted sparse nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3909–3921, Jul. 2017.

[13] M. Wang, B. Zhang, X. Pan, and S. Yang, "Group low-rank nonnegative matrix factorization with semantic regularizer for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 4, pp. 1022–1029, Apr. 2018.

[14] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.

[15] M. E. Winter, "N-FINDR: An algorithm for fast autonomous spectral end-member determination in hyperspectral data," in *Proc. Imag. Spectrometry V*, Denver, CO, USA, Oct. 1999, pp. 266–275.

[16] J. M. P. Nascimento and J. M. B. Dias, "Vertex component analysis: A fast algorithm to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 4, pp. 898–910, Apr. 2005.

[17] J. Li and J. M. Bioucas-Dias, "Minimum volume simplex analysis: A fast algorithm to unmix hyperspectral data," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Boston, MA, USA, vol. 3, Jul. 2008, pp. 250–253.

[18] C. H. Chen and X. Zhang, "Independent component analysis for remote sensing study," in *Proc. Image Signal Process. Remote Sens. V*, Florence, Italy, Dec. 1999, pp. 150–159.

[19] T.-M. Tu, "Unsupervised signature extraction and separation in hyperspectral images: A noise-adjusted fast independent component analysis approach," *Opt. Eng.*, vol. 39, no. 4, p. 897, Apr. 2000.

[20] V. P. Pauca, J. Piper, and R. J. Plemmons, "Nonnegative matrix factorization for spectral data analysis," *Linear Algebra Appl.*, vol. 416, no. 1, pp. 29–47, Jul. 2006.

[21] L. Miao and H. Qi, "Endmember extraction from highly mixed data using minimum volume constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 3, pp. 765–777, Mar. 2007.

[22] Y. Qian, S. Jia, J. Zhou, and A. Robles-Kelly, "Hyperspectral unmixing via $L_{1/2}$ sparsity-constrained nonnegative matrix factorization," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 11, pp. 4282–4297, Nov. 2011.

[23] X. Lu, H. Wu, Y. Yuan, P. Yan, and X. Li, "Manifold regularized sparse NMF for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 51, no. 5, pp. 2815–2826, May 2013.

[24] E. M. M. B. Ekanayake *et al.*, "Enhanced hyperspectral unmixing via non-negative matrix factorization incorporating the end member independence," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul./Apr. 2019, pp. 2256–2259.

[25] J. M. Bioucas-Dias *et al.*, "Hyperspectral unmixing overview: Geometrical, statistical, and sparse regression-based approaches," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 354–379, Apr. 2012.

[26] B. Rathnayake, K. M. K. Weerakoon, G. M. R. I. Godaliyadda, and M. P. B. Ekanayake, "Toward finding optimal source dictionaries for single channel music source separation using nonnegative matrix factorization," in *Proc. IEEE Symp. Ser. Comput. Intell. (SSCI)*, Bengaluru, India, Nov. 2018, pp. 1493–1500.

[27] P. O. Hoyer, "Non-negative matrix factorization with sparseness constraints," *J. Mach. Learn. Res.*, vol. 5, pp. 1457–1469, Dec. 2004.

[28] X. Liu, W. Xia, B. Wang, and L. Zhang, "An approach based on constrained nonnegative matrix factorization to unmix hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 2, pp. 757–772, Feb. 2011.

[29] J. Liu, J. Zhang, Y. Gao, C. Zhang, and Z. Li, "Enhancing spectral unmixing by local neighborhood weights," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 5, no. 5, pp. 1545–1552, Oct. 2012.

[30] L. Tong, J. Zhou, X. Bai, and Y. Gao, "Dual graph regularized NMF for hyperspectral unmixing," in *Proc. Int. Conf. Digit. Image Comput., Techn. Appl. (DICTA)*, Wollongong, NSW, Australia, Nov. 2014, pp. 1–8.

[31] Z. J. Xiang, Y. Wang, and P. J. Ramadge, "Screening tests for lasso problems," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 5, pp. 1008–1027, May 2017.

[32] J. Mairal, F. Bach, J. Ponce, G. Sapiro, and A. Zisserman, "Non-local sparse models for image restoration," in *Proc. IEEE 12th Int. Conf. Comput. Vis.*, Sep. 2009, pp. 2272–2279.

[33] J. Wright, A. Y. Yang, A. Ganesh, S. S. Sastry, and Y. Ma, "Robust face recognition via sparse representation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 31, no. 2, pp. 210–227, Feb. 2009.

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[34] K. Yu, T. Zhang, and Y. Gong, "Nonlinear learning using local coordinate coding," in *Advances in Neural Information Processing Systems 22*, Y. Bengio, D. Schuurmans, J. D. Lafferty, C. K. I. Williams, and A. Culotta, Eds. New York, NY, USA: Curran Associates, 2009, pp. 2223–2231. [Online]. Available: http://papers.nips.cc/paper/3875-nonlinear-learning-using-local-coordinate-coding.pdf

[35] T. N. Sainath, A. Carmi, D. Kanevsky, and B. Ramabhadran, "Bayesian compressive sensing for phonetic classification," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process.*, Mar. 2010, pp. 4370–4373.

[36] K. K. Chang, J.-S. R. Jang, and C. Iliopoulos, "Music genre classification via compressive sampling," in *Proc. 11th Int. Soc. Music Inf. Retr. Conf.*, 2010, pp. 387–392.

[37] F. Chen and Y. Zhang, "Sparse hyperspectral unmixing based on constrained l_p-l_2 optimization," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 5, pp. 1142–1146, Sep. 2013.

[38] J. Sigurdsson, M. O. Ulfarsson, and J. R. Sveinsson, "Hyperspectral unmixing with ℓ_q regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 11, pp. 6793–6806, Nov. 2014.

[39] C. Shi and L. Wang, "Incorporating spatial information in spectral unmixing: A review," *Remote Sens. Environ.*, vol. 149, pp. 70–87, Jun. 2014.

[40] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, May 2013.

[41] P. Di Lorenzo, P. Banelli, E. Isufi, S. Barbarossa, and G. Leus, "Adaptive graph signal processing: Algorithms and optimal sampling strategies," *IEEE Trans. Signal Process.*, vol. 66, no. 13, pp. 3584–3598, Jul. 2018.

[42] S. Segarra, A. G. Marques, and A. Ribeiro, "Optimal graph-filter design and applications to distributed linear network operators," *IEEE Trans. Signal Process.*, vol. 65, no. 15, pp. 4117–4131, Aug. 2017.

[43] X. Dong, D. Thanou, M. Rabbat, and P. Frossard, "Learning graphs from data: A signal representation perspective," *IEEE Signal Process. Mag.*, vol. 36, no. 3, pp. 44–63, May 2019.

[44] F. J. Iglesias, S. Segarra, S. Rey-Escudero, A. G. Marques, and D. Ramirez, "Demixing and blind deconvolution of graph-diffused sparse signals," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Calgary, AB, Canada, Apr. 2018, pp. 4189–4193.

[45] Y. Bai, G. Cheung, X. Liu, and W. Gao, "Graph-based blind image deblurring from a single photograph," *IEEE Trans. Image Process.*, vol. 28, no. 3, pp. 1404–1418, Mar. 2019.

[46] S. Grotas, Y. Yakoby, I. Gera, and T. Routtenberg, "Power systems topology and state estimation by graph blind source separation," *IEEE Trans. Signal Process.*, vol. 67, no. 8, pp. 2036–2051, Apr. 2019.

[47] L. Tong, J. Zhou, X. Li, Y. Qian, and Y. Gao, "Region-based structure preserving nonnegative matrix factorization for hyperspectral unmixing," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 4, pp. 1575–1588, Apr. 2017.

[48] R. Ammanouil, A. Ferrari, and C. Richard, "Hyperspectral data unmixing with graph-based regularization," in *Proc. 7th Workshop Hyperspectral Image Signal Process., Evol. Remote Sens. (WHISPERS)*, Jun. 2015, pp. 1–4.

[49] N. Gillis, "The why and how of nonnegative matrix factorization," in *Regularization, Optimization, Kernels, and Support Vector Machines*, vol. 12, no. 257. New York, NY, USA: CRC Press, 2014.

[50] D. D. Lee and H. S. Seung, "Algorithms for non-negative matrix factorization," in *Proc. Adv. Neural Inf. Process Syst.*, Denver, CO, USA, 2001, pp. 556–562.

[51] Z. Li, J. Chen, and S. Rahardja, "Graph construction for hyperspectral data unmixing," in *Hyperspectral Imaging in Agriculture, Food and Environment*. London, U.K.: IntechOpen, 2018.

[52] F. R. Chung and F. C. Graham, *Spectral Graph Theory*. Providence, RI, USA: American Mathematical Society, 1997.

[53] H. T. Nguyen and V. Kreinovich, *Applications of Continuous Mathematics to Computer Science*. Norwell, MA, USA: Kluwer, 1997.

[54] U. M. Ascher, E. Haber, and H. Huang, "On effective methods for implicit piecewise smooth surface recovery," *SIAM J. Sci. Comput.*, vol. 28, no. 1, pp. 339–358, Jan. 2006.

[55] V. Kreinovich and A. Neumaier, "For piecewise smooth signals, ℓ_1-method is the best among ℓ_p-methods: An interval-based justification of an empirical fact," *Comput. Technol.*, vol. 22, no. 2, pp. 50–58, 2017.

[56] W. J. Rugh, *Linear Systems Theory*, 2nd ed. Upper Saddle River, NJ, USA: Prentice-Hall, 1996.

[57] C.-I. Chang and Q. Du, "Estimation of number of spectrally distinct signal sources in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 3, pp. 608–619, Mar. 2004.

[58] J. M. Bioucas-Dias and J. M. P. Nascimento, "Hyperspectral subspace identification," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 8, pp. 2435–2445, Aug. 2008.

[59] D. C. Heinz and Chein-I-Chang, "Fully constrained least squares linear spectral mixture analysis method for material quantification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 39, no. 3, pp. 529–545, Mar. 2001.

[60] S. Jia and Y. Qian, "Spectral and spatial complexity-based hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 3867–3879, Dec. 2007.

[61] S. Jia and Y. Qian, "Constrained nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 1, pp. 161–173, Jan. 2009.

[62] F. Zhu, Y. Wang, S. Xiang, B. Fan, and C. Pan, "Structured sparse method for hyperspectral unmixing," *ISPRS J. Photogram. Remote Sens.*, vol. 88, pp. 101–118, Feb. 2014. [Online]. Available: http://www.sciencedirect.com/science/article/pii/S0924271613002761

[63] *Datasets & Ground Truths*. Accessed: Nov. 24, 2019. [Online]. Available: http://www.escience.cn/people/feiyunZHU/Dataset_GT.html

[64] R. A. Horn and C. R. Johnson, *Matrix Analysis*. New York, NY, USA: Cambridge Univ. Press, 1998.



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